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Robust and Efficient Claim Retrieval for Online Fact-Checking Applications

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Abstract

Understanding the veracity of statements is important when consuming information on the Web. Whereas fact-checking sites have provided a large corpus of already verified claims, matching a given utterance to already fact-checked claims remains a challenging task. Verified claim retrieval has been approached through a variety of different methods, among them approaches relying on supervised neural models. Whereas such models tend to perform strongly, they require significant training effort and their robustness towards unseen data distributions may vary heavily. Also, prior works demonstrate the capability of unsupervised models to provide state-of-the-art performance. In this paper, we assess established claim retrieval benchmark datasets and experimentally evaluate and compare different state-of-the-art supervised and unsupervised methods with regard to performance, but also computational effort

and run time. We show that unsupervised approaches outperform supervised ones with respect to robustness. While the best state-of-the-art method relies on supervised deep neural networks, its high computational costs make it difficult to use in online fact-checking applications. The best unsupervised method reaches a similar performance and meets efficiency requirements of online application scenarios due to low hardware requirements. Our experiments verify that, due to the nature of the task and data, the choice of pre-trained language models is more important than fine-tuning and that training supervised models on the target data may not be cost-efficient in online claim retrieval applications.

Keywords: Claim Retrieval, Fact-Checking, Machine Learning, Deep Learning, Fake News, Information Retrieval

1 Introduction

The spread of controversies, biased discourse, and falsehoods on the Web has become an increasingly important issue, from both a societal as well as a research perspective [1, 2]. False information has been shown to spread rapidly, and even faster than the truth [1]. Checking the veracity of information before it can spread is thus crucial. However, fact-checking is costly, requiring human expertise and time. As of now, over 100 fact-checking initiatives exist worldwide.¹ As manually fact-checking a single claim may take more than a week [3], one of the initial steps for fact-checkers is to find out if a claim has already been verified by another organization.

However, given that a particular claim proposition may occur in the form of diverse utterances [4], matching a given statement or utterance to fact-checked claims available from fact-checking portals remains a challenging problem. This problem is known as verified claim retrieval and has been recognised by the *Check That!* initiative [5–7], that has advanced research in this field by producing benchmarks and baselines and organizing shared tasks.

A variety of approaches has since been proposed. However, those were evaluated on different individual datasets, making it hard to compare their performance, their strengths, and weaknesses, and assess their robustness. Generally, while supervised approaches usually dominate regarding task performance, they tend to overfit the data distributions prevalent in the training corpus at hand, making them less robust on new data and evolved data distributions. Also, they are prone to picking up spurious correlations in the data, causing them to perform well on a given dataset rather than performing well on the task [8]. It is thus crucial to understand the characteristics of benchmark datasets, to compare the performance of methods across different datasets, and to understand how well the data reflects the task in the real-world.

¹<https://ifcncodeofprinciples.poynter.org/signatories>

Another important aspect that has so far achieved little attention is the question whether proposed approaches are actually useful in practice [9], in this case, to link unverified statements to a database of fact-checked claims, such as ClaimsKG [10, 11], as proposed by [12] with their ClaimLinker system. This means that beyond the quality of predictions, method performance with respect to run time and computational costs needs to be understood. This involves both efficiency during inference as well as offline computation costs during training, especially since dynamically emerging and evolving vocabulary has proven a significant challenge, requiring frequent updates of models to ensure stable performance over time [13]. This does not only pose challenges for practical reasons, but also leads to high energy consumption, exacerbating environmental issues [14], especially when unnecessarily large language models are applied [15].

With this work, we investigate the state-of-the-art in the retrieval of previously fact-checked claims, assessing used benchmark datasets and the performance of claim retrieval models. In particular, we aim at understanding whether the performance gains on particular datasets justify the required effort and computational and environmental costs for training and fine-tuning large supervised models.

We also compare the efficiency of approaches and their scalability. Given the growing amount of previously fact-checked claims and the importance of fast response times in real-world fact-checking scenarios, fast run time at inference is a crucial criterion.

To investigate these questions, we (a) assess established benchmark datasets for the retrieval of previously fact-checked claims to aid the interpretation of evaluation results, (b) compare 22 (15 supervised and 7 unsupervised) state-of-the-art methods with respect to performance, and (c) replicate and newly generate additional results on more datasets for 8 of these systems (4 supervised and 4 unsupervised) to compare them with respect to overall performance, robustness, scalability and run time.

We find that fine-tuning and supervision can indeed increase performance at the cost of risking overfitting. For instance, the overall best approach, *RIET Lab*, is supervised and relies on large language models. It scores between 0.008 and 0.039 points higher in MAP@5 on all datasets but one than the overall second best approach, unsupervised *SimBa 2023* which uses considerably smaller pre-trained models. However, *SimBa 2023* scores 0.108 points higher on the remaining dataset. The choice of pre-trained models and embeddings turns out to be more crucial than fine-tuning and the high computational costs for supervision may not be worth the performance gain for real applications, given that the performance difference between the best supervised approach and the best unsupervised approach averages to -0.07 points in MAP@5 over all datasets while the unsupervised approach achieves lower retrieval times on a laptop than the supervised approach on a high-performance GPU server, which usually is not available for online fact-checking applications.

The main contributions of our work are:

- an assessment of established benchmark datasets for the retrieval of previously fact-checked claims
- a comparative evaluation of different supervised and unsupervised approaches on established datasets for the claim retrieval task and extensive experimental results w.r.t. robustness of the approaches comparing the performances when training and testing on claims of different discourse types
- the assessment of the applicability of different approaches for real-world claim retrieval applications based on performance, efficiency and computational expenses
- modifications of existing approaches to allow cross-evaluation and application on different datasets. All modifications and all scripts to replicate this study are released to the community.

The paper is structured as follows. We start with the problem definition and introduction of our research questions (Section 2), before giving an overview of related work and the state of the art (Section 3). Section 4 details the compared claim retrieval approaches and our experimental setup. Our data analysis follows in Section 5 before we present and discuss the results of our comparative evaluation in Section 6. Section 7 concludes the paper with a discussion of future work.

The source code and detailed documentation and instructions to reproduce our results are available on [GitLab](https://git.gesis.org/bolandka/claimlinking)²

2 Problem Definition and Research Questions

From an application perspective, the task of retrieving previously fact-checked claims represents the initial step of fact-checking, e.g. in journalistic or news contexts, aiming at identifying if a given statement has been already fact-checked by trusted sources and finding fact-checked statements that are relevant to determining the veracity of the input claim.

More formally, given an input query q (e.g. the text of a tweet, or a sentence in a political debate) and a dataset D of previously fact-checked verified claims (with each verified claim denoted as c), the *claim retrieval* task aims at providing a ranked list of top- n claims C from D . Note here that a retrieved claim c consists of not only the text of the utterance but also additional metadata such as the title of the fact-checking article.

For example, consider the sentence from a political debate as input query: “*You know, interest on debt will soon exceed security spending.*”

Table 1 displays the top-3 list of previously fact-checked claims provided by the ClaimLinker [12] system.

In this context, the size of the dataset D (number of previously fact-checked claims) is important because it can affect both the retrieval effectiveness and the retrieval efficiency. In particular, a larger dataset might mean more semantically similar claims (which can make retrieval more difficult and affect

²<https://git.gesis.org/bolandka/claimlinking>

<i>c</i> text	<i>c</i> speaker	<i>c</i> date	<i>c</i> source	<i>c</i> ver- dict
Within a few years, we will be spending more on interest payments than on national security	Mitch Daniels	11 February 2011	politifact.com	false
In just 17 years, spending for Social Security, federal health care and interest on the debt will exceed all tax revenue	Dave Brat	29 May 2015	politifact.com	mostly true
By 2022, just the interest payment on our debt will be greater than the defense of our country	Joe Manchin	9 May 2018	politifact.com	mostly true

Table 1: Top-3 verified claims retrieved by the ClaimLinker system for the input query “*You know, interest on debt will soon exceed security spending.*”

precision) and higher response times, especially for supervised methods that often require pair-wise comparisons. Operating on a small dataset might lead to more unmatchable input claims, reducing a system’s recall.

In this paper, we investigate the following research questions:

1. RQ1: What are the characteristics of the task of retrieving previously fact-checked claims and existing benchmark datasets?
2. RQ2: Which supervised and unsupervised state-of-the-art approaches perform best on these established datasets and why?
3. RQ3: How robust is their performance depending on the training data?
4. RQ4: How efficient are the different approaches and how well do they scale depending on the number of verified claims in the dataset?
5. RQ5: Which approach is the best choice for (online) claim retrieval applications?

3 Related Work

In the following, we provide an overview of existing claim retrieval tasks, datasets, approaches and studies giving insights regarding dataset characteristics that are relevant to this work.

3.1 Claim retrieval tasks and datasets

A number of shared tasks and datasets have been proposed that evolve around fact-checking and the linking of input claims to information that can aid with fact-checking.

FEVER [16], for Fact Extraction and VERification, is a shared task focusing on checking factual claims. It consists of the retrieval of evidence from Wikipedia and using it to verify the claims. The FEVER dataset [17] consists of 185,445 claims generated by modifying Wikipedia excerpts that are subsequently verified independently from the original part of the text from

which they are derived. The contained claims and their corresponding evidence (sentences that support or refute them) are categorized into three classes: Supported, Refuted, or NotEnoughInfo.

Also dealing with the problem of claim verification and the categories Supports, Refutes, NoInfo/NotEnoughInfo, the dataset SciFact [18] and its recent extension SciFact Open [19] focus on claims and evidence from the scientific domain. Claims are created by human experts based on citing sentences. Corresponding evidence is drawn from the abstracts of the cited publications. SciFact contains 1.4k claims paired with annotated abstracts from a selection of the Semantic Scholar Open Research Corpus S2ORC [20] containing corresponding evidence. SciFact Open retrieves additional abstracts and evidence annotations for 279 of these claims, arriving at 500k research abstracts, 279 claims, and 460 claim-evidence pairs in the corpus. Automated methods are used to filter potentially relevant abstracts contained in the full S2ORC corpus for manual annotations.

The Fake News Challenge (FNC) [21] is about classifying the stance of news articles to claims (agrees, disagrees, discusses, unrelated). The benchmark data it provides consists of 75,385 labeled headline and article pairs, with the news article headlines serving as claims [22].

While these are related tasks, the linked evidence and articles cannot be used to directly infer veracity labels for a claim as is expected for verified claim retrieval. Rather, the contained evidence has to be reviewed and assessed by experts.

The MultiFC [23] and the ClaimsKG [10, 11] datasets both provide structured data of and about claims coming from reputed fact-checking portals, where claims are stored together with rich metadata (such as authors, sources, claim reviews and other contextual information, including veracity labels). While both datasets retrieve and store claims from fact-checking portals, they are complementary in a couple of aspects. MultiFC focuses on evidence-based fact-checking in terms of downstream tasks, where via the Google Search API the 10 most highly ranked search results per claim are retrieved and stored. ClaimsKG, on the other hand, provides a rich data model (an RDFS ontology) to represent check-worthy (or fact-checked) claims and related metadata, which is an important effort towards standardization and enables federated access to distributed data, where a specific search engine is provided³ in addition to a public Sparql endpoint [24]. MultiFC contains data in English, ClaimsKG is multilingual, harvesting data from fact-checking portals in English, French, Russian, Spanish, Italian, and Hindi. These datasets can be used to provide a pool of verified claims with additional metadata for fact-checking applications and to extract links to claims that are mentioned in fact-checking articles.

One initiative that directly focuses on the linking of unverified claims to previously fact-checked claims is the CheckThat! Lab [6, 7, 25]. From 2020 until 2022, its second task has been part of the CLEF campaign, providing a forum for development and evaluation of systems for the retrieval of verified

³<https://data.gesis.org/claimskg-explorer/home>

claims for tweets and, since 2021, statements from political debates. Here, the claim retrieval task is defined as *Given a check-worthy input claim and a set of verified claims, rank the previously verified claims in order of usefulness to fact-check the input claim* [6, 7] and *Given a check-worthy input claim and a set of verified claims, rank those verified claims, so that the claims that can help verify the input claim, or a sub-claim in it, are ranked above any claim that is not helpful to verify the input claim* [25, 26]. In 2020, the task has been coined *Verified Claim Retrieval*. In 2021 and 2022, it has been referred to as *Detecting Previously Fact-Checked Claims*. Both titles refer to the same ranking task. The task was offered for multiple languages. The CheckThat! Lab features real claims uttered by humans: Task 2A treats check-worthy tweets as input claims, Task 2B contains claims made in political debates or speeches. In both cases, real fact-checks from fact-checking portals are used to construct the set of verified claims to match against, partly drawn from ClaimsKG.

Since this task definition matches our problem definition, we choose the CheckThat! Lab datasets and the participating systems for our study.

3.2 Claim retrieval approaches

Early work has discussed the task of retrieving fact-checked claims as part of an end-to-end fact-checking pipeline, without providing detail or an evaluation [27]. Hence, the two ground-laying works in that field are [26] and [3]. In [26], the authors provide two datasets: one using Politifact articles and one using tweets mentioned in Snopes articles. The latter and the annotation protocol later became the basis for the CLEF CheckThat Lab! claim retrieval datasets. They evaluate a standard information retrieval method (BM25) and provide a definition of the task. In [3], the authors extend this method and incorporate global and local contextual information contained within the source texts, i.e. the political debate containing the input claims, and the target texts, i.e. the fact-checking articles containing the verified claims. They find that contextual information, especially on the source-side, helps the claim retrieval performance. A multi-modal approach was proposed by [28], whose neural ranking model uses both text and images for the retrieval and recommendation of claims about images, based on mining tweets containing images and links to fact-checking articles. On the same dataset and on data from the Chinese platform Weibo, [29] developed and evaluated their memory-enhanced transformer-based approach to select key sentences from fact-checking articles to estimate the relevance of a verified claim for a given input claim.

Systems submitted to the CheckThat! Lab shared claim retrieval tasks range from relying on pre-trained or fine-tuned language models and using supervised methods to efficient index-based systems relying on traditional information retrieval techniques such as BM25 in supervised or unsupervised setups. Some systems performed data augmentation and different data cleaning and pre-processing steps. The most successful approaches relied on transformers, achieving up to 0.956 MAP@5 on the 2022 English tweets dataset. We describe the systems in more detail in Section 4.

Several systems used the CheckThat Lab! benchmark without participating in the challenges. Building on the winner of the 2021 competition, [30] proposed a framework using distant supervision and a modified self-adaptive training approach to retrieve fact-checked claims for claims made in tweets. Their silver standard corpus is created by harvesting tweets that contain a link to a fact-check from Snopes, similar to [28]’s mining of tweets and fact-checking articles, yet focused on textual claims and implemented an automatic filtering and labeling step. They beat the state-of-the-art performance on the CLEF CheckThat! Lab 2021 English tweet data (0.882) with a MAP@5 score of 0.903. The web-based online claim retrieval tool ClaimLinker [12] links arbitrary texts to claims in ClaimsKG. It obtains candidate claims through a blocking step based on Elasticsearch’s (ES) BM25 scores and re-ranks them using textual similarity measures. Despite its simplicity, it succeeds in retrieving the correct verified claim in the first position (P@1) in 76% of the cases for the CheckThat! Lab tweets dataset of 2020 (83.5% for P@5). MAP scores are not specified by the authors. [31] preprocess tweets and enrich them with extracted information from embedded URLs, images, and videos. They experiment with different claim numbers for a BM25-based blocking step and different BERT variants for reranking the results. Their best approaches beat the previous state-of-the-art performance on CheckThat! Lab English tweet data of the years 2020 and 2021 with MAP@5 scores of 0.955 and 0.936, respectively. However, the 2021 system has access to additional metadata. Using the same system as for 2020, the score on 2021 drops to 0.929. This shows that the datasets of the different years have different characteristics or task difficulties, which makes the different systems proposed in the literature hard to compare, even when they use the CheckThat! Lab benchmark and the same evaluation metrics. We will provide more insights both into the differences of the datasets and the performances of the systems in Sections 5 and 6.

3.3 CheckThat! Lab claim retrieval data analyses

Several works investigate characteristics of claim retrieval benchmark datasets that are relevant to assess task difficulty, robustness of methods, and possible biases during evaluation.

[26] include an exploratory analysis of their dataset, giving examples of trivial cases where the verified claim is identical to the input claim, harder cases where they differ regarding lexical choice, and complex input claims with sub-claims, which are verified by two corresponding verified claims. They further distinguish input claim and verified claim pairs into two categories: those that can be matched using simple approximate string matching techniques (Type-1), and all others (Type-2). Analyzing a manually annotated sample of 100 pairs extracted from their politifact debates dataset, they found 48% to be of Type-2. An analysis of word-level TF.IDF-weighted cosine-similarities further revealed their politifact debates data to be harder than their tweet data from Snopes, with 27% of the debates pairs having similarity scores above 0.25 vs. 50% of the tweets data. [3] analyzed the debates dataset by [26] in more detail

and categorized the pairs of matching claims into 4 categories: (1) clean, (2) clean-hard, (3) part-of and (4) context-dep. (1) and (2) refer to self-contained input claims with a matching verified claim that directly (1) vs. indirectly (2) verifies it, (3) and (4) refer to input claims that are not self-contained and require additional sentences from the discursive context to fully form an individual claim (3) or need co-reference resolution to contain all relevant information (4). Furthermore, the authors show that different data splits can greatly influence task difficulty because systems can benefit from training and testing on claims drawn from the same debates or from debates about similar topics.

While [26]’s data and annotation protocol later formed the basis of the CheckThat! Lab datasets, the latter contained additional data and different splits, thus the results cannot be transferred to these benchmark datasets. However, they highlight the importance of analyzing the data in more detail to understand and compare model performance.

[32] analyzed the number of words and sentences in the different fields for the CheckThat! Lab 2021 data. [30] provide insights on similarities between input tweets and titles and subtitles of matching verified claims, compare Jac-card vs. Cosine similarity scores, and analyze Sentence-BERT-based retrieval performance for different similarity thresholds.

While these analyses add valuable insights for understanding the performance of their models, a comprehensive analysis across datasets and the relation of dataset characteristics and performance of different models is hitherto missing. We aim to close this gap with this study.

4 Experimental Setup

In the following, we describe our evaluation protocol including analyzed datasets, evaluated methods, and evaluation metrics.

4.1 Datasets

For comparative evaluation, we use the datasets and corresponding training/test splits as provided by CheckThat! Lab Task 2 (*verified claim retrieval*) challenges of the CLEF2020 [25], CLEF2021 [6] and CLEF2022 [7] initiatives in English language. Table 2 provides statistics regarding training and test data for each dataset. The dataset from the first edition (in the following referred to as *2020-tweets*) contains claims extracted from tweets as input queries and corresponding verified claims from fact-checking websites as targets. The targets were drawn from the fact-checking platform Snopes, harvested and augmented with data from ClaimsKG [10] as described in [25] and [26]. The data from the 2021 and 2022 challenges each contain one dataset with tweets (Task 2A, in the following referred to as *2021-tweets* / *2022-tweets*), extensions of the previous datasets, and one with a different discourse type: claims from political debates (Task 2B, referred to as *2021-debates* / *2022-debates*).

Claims from political debates were mainly drawn from the fact-checking portal Politifact using the same protocol as described in [26]. The ground truth of links between input queries and matching verified claims was constructed using information provided by Politifact overview articles which feature links between claims from political speeches and fact-checks on their portal, and by Snopes, where the review articles often refer to tweets spreading the fact-checked claim.

For the tweet data, there is always exactly one correct verified claim to retrieve per input query. For claims from political debates, the number of correct matches may be greater than one. A verified claim may be the correct target for more than one input query in all datasets. Each dataset contains a set of target verified claims to match all queries against: queries from test, training, and development sets alike. I.e. there always is one set of target verified claims to use both for training and testing. The data for each year consists of the data of the previous year with some additions. The test sets, consisting of a file with input queries, i.e. their IDs and texts, and a file with gold matchings between input query IDs and target verified claim IDs, contain new input queries each year and links to either new targets or links to verified claims that have already been contained in the set of verified claims in previous editions. The training sets also consist of an input query file and a file with matchings between queries and targets. These files did not change after the 2021 edition, i.e. the training sets of *2021-tweets* and *2022-tweets* are the same, and the training sets of *2021-debates* and *2022-debates* are the same. However, the set of target verified claims to match against differs for *2021-tweets* vs. *2022-tweets*, i.e. the latter contains additional candidate claims. Thus, training a model on data from the respective years may yield different results because a model has access to additional negative examples during training. The set of verified claims is the same for *2021-debates* and *2022-debates*. This means that only new input claims were added for the test set, but no new matching verified claims, i.e. the new input queries have links to targets that have already been contained in the set of verified claims in the earlier years. Test sets of a previous year were used as development sets for the upcoming year.

Since there always is one set of target verified claims to use both for training and testing, methods use the set of verified claims that is also used for testing when training and fine-tuning, but paired with different input queries.

The debates data offers fewer positive training examples, more candidates to choose from, and requires finding more than one match for some input claims, which supposedly makes it more difficult than the tweet data.

We investigate the characteristics of the data in more detail in Section 5 to address RQ1.

4.2 Retrieval performance and robustness

For investigating RQ2, we train and test all available methods on all datasets for a comparative evaluation. For RQ3, we perform a cross-dataset evaluation: we apply the trained models to test data of different datasets. This serves 1) to

Name	#ITr	#VTr	#ITe	#VTe	#VClaims
2020-tweets	800	800	200	200	10,375
2021-tweets	999	999	202	202	13,825*
2022-tweets	999	999	209	209	14,231
2021-debates	472	562	79*	103	19,250
2022-debates	472	562	65*	83*	19,250*

Table 2: Datasets and statistics. *#ITr*: Number of input claims in the training set, *#VTr*: Number of input claim - verified claim pairs in the training set, *#ITe*: Number of input claims in the test set, *#VTe*: Number of input claim - verified claim pairs in the test set, *#VClaims*: Total number of verified claims to match against. *Numbers based on our own counts, diverge from the numbers specified in the task overview paper

analyze the influence of the training sets without confounding the results with potential differences in test set difficulty and 2) to assess the robustness of the methods and how well-learned models generalize to data of different distributions and of a different discourse type (tweets vs. claims from political debates). More precisely, we train supervised approaches on each of the *2020-tweets*, *2021-tweets*, *2022-tweets*, and *2021/2-debates* training datasets and apply each of the trained models on all of the test datasets. Since *2021-debates* and *2022-debates* share the same training sets and verified claims, we only train one model per approach on the debates data. We train three models for each neural network-based approach on each dataset to investigate their robustness/stability and report the average scores. With this, we compare performances of the supervised approaches for in-distribution vs. out-distribution training and relate these scores with the performance of unsupervised approaches on each test dataset.

We rely on the performance metrics used in the official CLEF CheckThat! Lab challenges: Mean Average Precision@k (MAP@k) and Precision@k with MAP@5 being the primary measure to rank the performance of systems [7]. Note that these metrics are applicable both for the single and the multiple target case.

4.3 Efficiency

For investigating RQ4, efficiency, and scalability, we analyze runtime complexity and costs for online computations of the different approaches. Online processing encompasses all steps that depend on the input claims and can only be executed at inference time: pre-processing and feature generation for input claims, comparisons with verified claims, and inference. Offline processing refers to pre-processing of verified claims in the database, e.g. computing and storing their embeddings or other features, and includes all training and fine-tuning steps. In order to assess applicability in real applications, we measure the time required for the offline and online processing steps of the best supervised and the best unsupervised approach in our comparison.

We use two different servers: *Server 1* with 4x NVidia GeForce RTX2080Ti (11 GB) GPU, 2 x Intel Xeon 2.1 GHz CPU with 24 available cores and 1.4 TB RAM, and *Server 2* with 1x A6000 GPU (48 GB) and Xeon E5-4667 v3 CPU with 32 available cores and 64 GB RAM. In addition, we examine whether the best unsupervised approach can be run on a laptop. We use a model with an AMD Ryzen 9 6900HS processor with 3.30 GHz Radeon Graphics and 16 GB RAM. We measure the average time needed to link an input claim to a claim in a database depending on the size of the database, i.e. the number of candidate verified claims. We compare the results for database sizes of 1k, 5k vs. 10k claims respectively. The respective samples were drawn randomly from the 2020-tweets dataset with all claims from the smaller sets being included in the larger ones. We do not store and use any intermediate results such as embeddings of already considered input claims and treat all experiments with the different database sizes independently. We measure three runs for each database size and report the averaged run times.

The number of input claims remains constant: we use the 200 input claims from the 2020-tweets test dataset.

4.4 Claim retrieval methods and baselines

In the following, we introduce the methods compared in this study: systems and baselines that were part of any of the CLEF CheckThat! Lab claim retrieval challenges (2020, 2021, and 2022) plus the online claim retrieval tool *Claim-Linker*, and an extension of the 2022 participant *SimBa*, *SimBa 2023*. We include here all systems for which publications or working notes are available.

An overview of the systems and their characteristics can be found in Table 3. While many systems experiment with the use of different verified claim fields, most use only the texts of the verified claims or the texts plus the titles of the fact-checking articles for prediction. Likewise, some systems experimented with incorporating the context information from the debates transcripts but none used them in their final submissions.

The individual systems are summarized below.

⁹<https://huggingface.co/sentence-transformers/bert-base-nli-mean-tokens>

¹⁰<https://huggingface.co/sentence-transformers/bert-large-nli-max-tokens>

¹¹<https://huggingface.co/sentence-transformers/distiluse-base-multilingual-cased-v1>

¹²<https://huggingface.co/sentence-transformers/all-mpnet-base-v2>

¹³<https://huggingface.co/princeton-nlp/sup-simcse-roberta-large>

¹⁴<https://github.com/facebookresearch/InferSent>

¹⁵<https://tfhub.dev/google/universal-sentence-encoder/4>

¹⁶<https://huggingface.co/sentence-transformers/all-mpnet-base-v2>

¹⁷<https://huggingface.co/princeton-nlp/sup-simcse-roberta-large>

¹⁸<https://github.com/facebookresearch/InferSent>

¹⁹<https://tfhub.dev/google/universal-sentence-encoder/4>

²⁰<https://huggingface.co/sentence-transformers/all-mpnet-base-v2>

²¹<https://huggingface.co/princeton-nlp/unsup-simcse-roberta-base>

²²<https://huggingface.co/sentence-transformers/sentence-t5-base>

²³<https://huggingface.co/sentence-transformers/sentence-t5-large>

²⁴<https://huggingface.co/EleutherAI/gpt-neo-1.3B>

²⁵<https://huggingface.co/sentence-transformers/bert-base-nli-mean-tokens>

²⁶<https://huggingface.co/bert-base-uncased>

Buster.ai [33] incorporates external data from similar tasks to fine-tune a pre-trained version of RoBERTa. They further clean all tweets and while training, focus on retrieving adversarial negative examples using indexed search to better distinguish negative examples with high similarities.

UNIPi-NLE [34] bases its predictions on a blocking step using an Elasticsearch (ES) index and word matching techniques and subsequent application of Sentence-BERT models fine-tuned in two steps.

UB_ET [35] retrieves the top 1k tweet–claim pairs using parameter-free DPH divergence from the randomness term weighting model in Terrier, computes several features from weighting models (BM25, PL2, and TF-IDF) and then builds a LambdaMart model on top for reranking.

Aschern [36] uses a fine tuned-sentence BERT model for blocking, then re-ranks the top 20 candidates using a trained LambdaMART reranker with Sentence-BERT and TF.IDF similarity scores.

CrowdChecked [30] adopts Aschern’s pipeline and adds modified self-adaptive training and distant supervision using claims and fact-checking URLs mined from social media.

NLytics [37] used similarities derived from a pre-trained RoBERTa model to rank the candidate target claims. This approach can thus be seen as a baseline for using RoBERTa without further additions.

DIPS [38] computes the cosine similarities for each input and verified claim pair using Sentence-BERT embeddings for the fields claim text, review title, review subtitle, and review date, and passes them as a sorted list to a neural network. For the debates data, they use the fields title and text.

BeaSkU [39] use a triplet loss training method to perform fine-tuning of the Sentence-BERT model and use the scores predicted by that model together with BM25 scores as features to train a rankSVM-based reranker.

bigIR did not submit a working note. However, their approach is described in detail in [31], and according to [7], the submitted variant retrieves candidates using BM25 on preprocessed and augmented tweets, as described in Section 3, and performs reranking using a fine-tuned MPNet model.

Fraunhofer SIT [40] augment the data using back-translation of tweet texts, train classifiers based on cosine-similarities of embeddings derived from fine-tuned sentence transformers for claim texts, review titles and subtitles, respectively, and combine the scores using an SVM as meta classifier. The authors accidentally submitted results generated by averaging the scores instead of using the better-performing meta classifier. We provide the results of their best-performing latter variant (Fraunhofer SIT SVM) in addition to the submitted variant.

For all publicly available submissions, detailed in the following, we additionally provide detailed cross-evaluation experiments and replicate their results. We modified the approaches to allow training and testing on all different datasets. All modified code can be found in our repository. Unless stated otherwise, we used the variants that constituted the teams’ primary submissions.

TIET [41] retrieves the top 1,000 claims for an input tweet based on their BM25 scores and combines those with a similarity score based on the cosine between Sentence-BERT (Base NLI Mean Tokens) embeddings for input and candidate claims.

NLP&IR@UNED [42] uses Universal Sentence Encoder [43] embeddings for input tweets and verified claims and train a feed-forward neural network with cosine similarity of embeddings, type-token ratio, average word length, number of verbs/nouns, ratio of content words, and ratio of content tags as features.

TheUofSheffield [44] is a point-wise Learning-to-Rank approach that trains a Linear SVM to learn scoring models based on BM25 scores and TF-IDF term weightings for claim texts and titles. Preprocessing consists of removing URLs, emojis, punctuation, and stopwords, converting to lowercase, tokenization, and Porter stemming and lemmatization.

Check_square [45] builds a KD-tree based on embeddings for claim texts and titles to retrieve the 1,000 most similar verified claims for an input claim. We report results for two of their variants which performed better than their primary submission: one using Sentence-BERT-Large pre-trained on SNLI with MAX tokens, fine-tuned with triplet loss, and the other one using multi-lingual DistilBERT-embeddings without fine-tuning (dismult). In both cases, the tweet pre-processing includes removing URLs, emails, phone numbers, and user mentions.

ES Baseline²⁷ retrieves claims from an ES index purely based on their BM25 scores. To maintain a consistent baseline, we utilized the script of 2021 and built the ES-index for the verified claims of the 2020 edition, which were presented in a different format, as described in the baseline script of 2020²⁸.

RIET Lab [46] uses a fine-tuned Sentence-T5 transformer model for blocking and a fine-tuned generative GPT-Neo model for re-ranking.

SimBa [47] uses an embedding-based blocking step followed by re-ranking based on the combination of multiple embeddings and lexical features. The 2022 challenge submission for the tweets data, **SimBa (tweets)** was unsupervised and combined the cosine similarity scores from the embeddings produced by all-mpnet-base-v2²⁹[48], SimCSE³⁰[49] and a token overlap count for re-ranking. Blocking was performed by selecting the union of the top 50 verified claims identified by either of the *all-mpnet-base-v2*, SimCSE, InferSent³¹[50], and Universal Sentence Encoder³²[51] embeddings. For the debates data, a supervised approach was used, **SimBa (debates)**. It used the blocking mechanism to select hard negatives for training a Linear SVC with random

²⁷https://gitlab.com/checkthat_lab/clef2021-checkthat-lab/-/blob/master/task2/baselines/elasticsearch_baseline.py

²⁸https://github.com/sshaar/clef2020-factchecking-task2/blob/master/elastic_search_baseline.py

²⁹<https://huggingface.co/sentence-transformers/all-mpnet-base-v2>

³⁰<https://huggingface.co/princeton-nlp/sup-simcse-roberta-large>

³¹<https://github.com/facebookresearch/InferSent>

³²<https://tfhub.dev/google/universal-sentence-encoder/4>

undersampling. As features, it relied on SimCSE embeddings, a string distance measure, token overlap, token overlap ratio, and overlaps and overlap ratios of WordNet synsets. We modify this approach, **SimBa 2023**, to use *all-mpnet-base-v2* for blocking and the Braycurtis distance³³ as a similarity measure, since it provided more accurate results than cosine similarity. For re-ranking, it uses *all-mpnet-base-v2*, SimCSE³⁴, Sentence-T5³⁵[52], Universal Sentence Encoder and the ratio of common words (without preprocessing).

Finally, we include the web-based online claim retrieval tool **ClaimLinker** [12] in our analyses. ClaimLinker obtains candidate claims through a blocking step based on BM25 scores and re-ranks them using textual similarity measures. For the variant reported here, we added preprocessing to remove special characters, punctuation, and stopwords, split hashtags, and lemmatize all claims using the Stanford CoreNLP Lemmatizer³⁶. This step aims to enable the tool to better process tweets. It uses text overlap features (common words, lemmas, entities, words+POS, ngrams, nchagrams) and combines BM25 and textual similarity scores with a weighting of 60/40.

³³<https://docs.scipy.org/doc/scipy/reference/generated/scipy.spatial.distance.braycurtis.html>

³⁴<https://huggingface.co/princeton-nlp/unsup-simcse-roberta-base>

³⁵<https://huggingface.co/sentence-transformers/sentence-t5-base>

³⁶<https://stanfordnlp.github.io/CoreNLP/lemma.html>

Approach	Su.	F.t.	Block	Idx	LMs	Fields
NLP&IR@UNED TIET	✓		1000	✓	bert-base-nli-mean-tokens ⁴	vclaim vclaim
TheUofSheffield ES-Baseline Check_square	✓		1000	✓	bert-large-nli-max-tokens ⁵	vclaim, title vclaim, title vclaim, title
Check_square dismult		✓	1000		distiluse-base-multilingual-cased ⁶	vclaim, title
ClaimLinker SimBa (tweets)			30 ≥50	✓	all-mpnet-base-v2 ⁷ , usup-simcse-roberta-large ⁸ , InferSent fastText ⁹ , UniversalSE ¹⁰	vclaim, title vclaim, title
SimBa (debates)	✓		≥50		all-mpnet-base-v2 ¹¹ , usup-simcse-roberta-large ¹² , InferSent fastText ¹³ , UniversalSE ¹⁴	vclaim, title
SimBa 2023			50		all-mpnet-base-v2 ¹⁵ , unsup-simcse-roberta-base ¹⁶ , sentence-t5-base ¹⁷	vclaim, title
RIET Lab	✓	✓	25		sentence-t5-large ¹⁸ , gpt-neo-1.3B ¹⁹	vclaim
Buster.ai	✓				RoBERTa*	augmented vclaims
UB.ET	✓		1000	✓		vclaim
UNIP1-NLE	✓	✓	1000	✓	bert-base-nli-mean-tokens ²⁰ , bert-base-uncased ²¹	vclaim, title
CrowdChecked	✓	✓	100		stsb-bert-base ²²	vclaim, title, subtitle
Aschern	✓	✓	20		stsb-bert-large ²³	vclaim, title, subtitle
NLytics DIPS (tweets)	✓		50		roberta-base ²⁴ paraphrase-distilroberta-base-v1 ²⁵	vclaim vclaim, title, subtitle, date
DIPS (debates)	✓		100		paraphrase-distilroberta-base-v1	title, text, query_ID: date
BeaSku	✓	✓			distilbert-base-nli-mean-tokens	vclaim, title, body
BigIR		✓	20	✓	STSB-MPNet-base ²⁶	vclaim, title
Fraunhofer SIT		✓			all-MiniLM-L6-v2, all-MPNet-Base-v2	vclaim, title, subtitle
Fraunhofer SIT SVM	✓	✓			all-MiniLM-L6-v2, all-MPNet-Base-v2	vclaim, title

Table 3: Claim retrieval approaches compared in this study. Su.: approaches using a supervised classifier or ranker. F.t.: approaches fine-tuning language models for the task. Block: Amount of candidates after the blocking step, if any. Idx: Approaches making use of an index such as Elasticsearch. LMs: used language models. *precise variant not specified. Fields: Verified target claim fields considered by the system: vclaim: text of the verified claim, (sub)title/text: (sub)title/text of the verified claim review article. Approaches above the dividing line are considered for additional cross-evaluation experiments.

5 Data Analysis

In order to facilitate an informed interpretation of experimental results of state-of-the-art systems and the utilised features, understanding the characteristics of the used datasets is crucial. This is even more important when aiming to understand model robustness and overfitting through cross-dataset evaluation. Thus, in the following, we provide an assessment of the aforementioned widely used benchmark datasets for claim retrieval evaluation.

We draw on characteristics investigated and deemed relevant for the task in previous works (cf. Section 4), such as claim and input lengths and token overlaps, and provide extended analyses of these properties for all different datasets, including statistics for inputs and verified claims in test vs. training splits. We add detailed insights into the completeness and correctness of the data and gold data annotations and their match to the task definition and real-world claim retrieval requirements.

5.1 Matches per input query and per verified claim

As per task definition, each input query in the tweets datasets has one matching verified claim in D . However, we find that in *2020-tweets*, one training query has two matching verified claims. In the *2021/2-tweets* training set, this is the case for two input queries: the query corresponding to the *2020-tweets* query plus a new entry. Moreover, there are duplicates in some of the gold input query - matching target verified claim pair files for the test datasets, as detailed in Table 4. Duplicates here mean repeated entries of the same input query - verified claim pair. Thus, the respective input queries implicitly gain a higher weight during evaluation than the ones that are not repeated. There are no duplicates in the input query files. For *2020-tweets*, one test input query (ID 1198) does not have a match to a verified claim in the gold file. Instead, there is a duplicate for another input query (ID 1167). A closer examination of the data reveals that the duplicated line can be corrected by changing the listed query ID with the missing query ID, i.e. both match the same verified claim (see below for statistics regarding multiple matches per verified claim). All other input queries are contained in the gold files. The other tweet datasets do not contain duplicates. However, as Table 4 shows, there are many duplicates in the debates test sets. For the debates data, 12.288% of training queries have more than one matching verified claim. For the test queries, this is the case for 16.456% (29.114% without de-duplication) for *2021-debates* and 18.462% (26.154% without de-duplication) for *2022-debates*. The average number of matching verified claims per test query remained stable: 1.165 (1.304 without de-duplication) for *2021-debates* vs. 1.185 (1.277 without de-duplication) for *2022-debates*. For training queries, the ratio is 1.191 (1.191 without de-duplication). All following analyses are performed on the original datasets, without de-duplication, to allow insights into the results reported in the literature which used the data as-is.

For all datasets, one verified claim may be the correct match for multiple input queries. Table 4 details the distribution for each dataset and split. For all datasets, the number of positive examples is low. The debates datasets contain fewer positive examples than the tweets datasets. When a verified claim is the correct match for an input query, it in many cases also matches other input queries (~ 18 - 20% for tweets training datasets, ~ 12 - 28% for tweets test, $\sim 36\%$ for debates training, ~ 37 - 44% for debates test). This is pronounced for the debates data. Therefore, the number of verified claims to choose for matching is low, as detailed in column $\#c_match$, and models may learn to identify relevant verified claims and that each of them has a high probability of matching multiple input queries.

dataset	split	$\#V$	$\#c$	$\#c_match$	$\#c_multi$
2020-tweets	train	801 (801)	10375	657 (6.333%)	115 (17.504%)
2020-tweets	test	200 (199)	10375	150 (1.446%)	42 (28.000%)
2021-tweets	train	999 (999)	13825	782 (5.656%)	159 (20.332%)
2021-tweets	test	202 (202)	13825	143 (1.034%)	31 (21.678%)
2022-tweets	train	999 (999)	14231	782 (5.495%)	159 (20.332%)
2022-tweets	test	209 (209)	14231	180 (1.265%)	21 (11.667%)
2021-debates	train	562 (562)	19250	354 (1.839%)	129 (36.441%)
2021-debates	test	103 (92)	19250	52 (0.270%)	23 (44.231%)
2022-debates	train	562 (562)	19250	354 (1.839%)	129 (36.441%)
2022-debates	test	83 (77)	19250	52 (0.270%)	19 (36.538%)

Table 4: Correct matches per verified claim. $\#V$: number of gold verified-input query pairs in total (and without duplicates), $\#c_match$: number of claims in the verified claims file that are the correct match for at least one input query (absolute and relative to $\#c$). $\#vc_multi$: number of claims in the verified claims file that are the correct match for more than one input query (absolute and relative to $\#c_match$)

In eight cases, a verified claim is the correct target for input queries in the training set and input queries in the test set, i.e. a model may learn to find matching claims for this specific verified claim in the training phase. This is true for seven verified claims in *2020-tweets* and one claim in *2022-tweets*.

5.2 Tweet datasets

For the tweets data, training, and test input queries consist of tweet texts including links to pictures or other URLs, user mentions, hashtags, and all other content allowed in tweets. In addition, they have the user name and mention as well as the date of publication appended to their end, separated from the tweet text by ”-”. For an example, see Table 5. The query with q_id 2 is one of three queries in the *2020-tweets* training set that is missing the

q_id	q	c_id	c_text	c_title
1	Trump needs to immediately divest from his businesses and comply with the emoluments clause. Iran could threaten Trump hotels *worldwide* and he could provoke war over the loss of revenue from skittish guests. His business interests should not be driving military decisions. — Ilhan Omar (@IlhanMN) January 6, 2020	394	In January 2020, U.S. Rep. Ilhan Omar advised Iran to attack Trump-branded hotels in the world, thus committing treason.	No, U.S. Rep. Ilhan Omar Didn't Give 'Treasonous' Military Advice to Iran
2	A number of fraudulent text messages informing individuals they have been selected for a military draft have circulated throughout the country this week.	670	The U.S. Army is sending text messages informing people they've been selected for the military draft.	Is US Army Sending Texts About a Military Draft?
3	Fact check: The U.S. Army is NOT contacting anyone regarding the draft. If you are receiving texts, phone calls or direct messages about a military draft, they are not official communications from the U.S. Army pic.twitter.com/3S32De8ekP — U.S. Army CGSC (@USACGSC) January 8, 2020	670	The U.S. Army is sending text messages informing people they've been selected for the military draft.	Is US Army Sending Texts About a Military Draft?

Table 5: Matching queries and targets from the 2020-tweets training dataset.

appended information. The *2021-tweets* training set misses this information in four cases. In all other data, the appended information is present. Therefore, *check-worthy claim* in this task refers to a tweet, which may consist of multiple sentences and a user mention and publication date.

Target verified claims of the *2020-tweets* data consist of the target ID, the text of the verified claim, and the title of the fact-checking article, as illustrated in Table 5.

In the 2021 and 2022 editions, the target verified claim data include further information from the fact-checks: review title, subtitle, date, text, author, and only in 2022, url of the review (in JSON format), see Figure 1.

As [38] observed, the appended date format of the input queries corresponds to the format of the date field in the target verified claims which is supplied for the 2021 and 2022 data, but not for 2020.

Concerning completeness of the verified claim data, the *2021-tweets* and *2022-tweets* data contain some empty or missing fields: subtitles are non-empty for all but 4 claims, authors for all but 26, dates for all but 287. The URLs in *2022-tweets* are only available for 2,266 of the 14,231 target verified claims. All other metadata is available for every target claim. In 9 cases, a *2020-tweets* target claim included a newline which, during conversion to JSON from tsv for

```
{
  'vclaim_id': 'vclaim-sno-15-homeless-dead-in-chicago',
  'vclaim': 'Fifteen homeless people in Chicago
were found dead on the street because of record-low temperatures that hit
the city in late January 2019.',
  'title': 'Were 15 Homeless People Found Frozen to Death in Chicago?',
  'subtitle': 'A picture of a Canadian man was repurposed online to prod
social media users to think about how "blessed" they are.',
  'page_url': 'https://www.snopes.com/fact-check/15-homeless-dead-in-chicago/',
  'author': 'Arturo Garcia',
  'date': '30 January 2019'
}
```

Fig. 1: Target verified claim in 2022-tweets

```
8935    The Monsanto Protection Act creates a \precedent-setting
limitation on\njudicial review of genetically-engineered crops."
Monsanto Protection Act
```

```
{
  'title': 'Monsanto Protection Act',
  'subtitle': "Does the Monsanto Protection Act create a 'precedent-setting
limitation on judicial review of genetically-engineered crops'?",
  'author': 'David Mikkelsen',
  'date': '13 September 2013',
  'vclaim_id': 'vclaim-sno-monsanto-protection-act',
  'vclaim': 'The Monsanto Protection Act creates a \precedent-setting
limitation on'
}
```

Fig. 2: The same target verified claim in the *2020-tweets* (top) vs. *2021/2-tweets* (bottom) datasets

the *2021-tweets* and *2022-tweets* datasets, caused the claim to be truncated at the newline, see Figure 2.

The input claims in test and training sets are ordered and their IDs are constructed depending on this order. Thus, the IDs provide useful information: claims matching the same target are grouped together, e.g. the *2020-tweets* test input queries with the IDs 999 and 1000 both match the verified claim with ID 6094. This is true for all datasets. This fact can potentially be used by supervised models.

5.3 Political debates datasets

For the debates data, the target verified claims consist of verified claim text (vclaim) and metadata from fact-checking review articles: URL, speaker, truth label, date, title and review text (truncated here), as shown in Figures 3 and 4.

```
{
"url": "/factchecks/2020/jan/15/elizabeth-warren/democratic-debates-biggest-
electoral-losers-number/",
"speaker": "Elizabeth Warren",
"vclaim": "\"Look at the men on this stage. Collectively, they have lost 10
elections. The only people on this stage who have won every single election
that they\u2019ve been in are the women, Amy and me.\\"",
"truth_label": "true",
"date": "stated on January 14, 2020 in a debate:",
"title": "The Democratic debates\u2019 biggest (electoral) losers, by the
numbers",
"text": <full review text>,
"vclaim_id": "vclaim-pol-17583"
}
```

Fig. 3: Target verified claim with ID *vclaim-pol-17583* from the *2021-debates* test dataset

```
{
"url": "/factchecks/2016/jan/13/barack-obama/business-has-created-jobs-
every-month-obamacare-be/",
"speaker": "Barack Obama",
"vclaim": "\"Our businesses have created jobs every single month since
(Obamacare) became law.\\"",
"truth_label": "true",
"date": "stated on January 12, 2016 in his State of the Union speech:",
"title": "Business has created jobs every month since Obamacare became
law, Obama said in State of the Union",
"text": <full review text>,
"vclaim_id": "vclaim-pol-05563"
}
```

Fig. 4: Target verified claim with ID *vclaim-pol-05563* from the *2021-debates* test dataset

Note that the speaker field may contain all kinds of information about a source of a fact-checked claim: it can be the actual speaker of an utterance, as in the example above, or other source information such as “Facebook posts”.

The test datasets include transcripts of the speeches, containing not only the text of the input claim utterances but also their discursive contexts. The input claims themselves are individual sentences from the speeches, yet they sometimes refer to entities mentioned in the context rather than the claims themselves, and can depend on one another, as can be seen in Table 6. With this, the data can be seen as diverging from the task definition defining *a check-worthy claim* as input, as the individual queries in this dataset are not necessarily check-worthy by themselves and do not constitute individual claims, depending on the precise definition of what a *claim* is (see [4] for an overview of diverging definitions), which is left implicit in this task definition.

q_id	q	c_id
2020_jan_15_441	Look at the men on this stage.	vclaim-pol-17583
2020_jan_15_442	Collectively, they have lost ten elections.	vclaim-pol-17583
20160112_Obama_ state_of_the_union_0013	And our businesses have created jobs every single month since it became law.	vclaim-pol-05563

Table 6: Input queries and matching target verified claims from the *2021-debates* test (rows 1-2) and training (row 3) sets. The verified claim targets are displayed in full in Figures 3 and 4

In both the 2021 and 2022 datasets, the input query IDs contain the date of the debate which can be used as a feature to compare with the verified claim dates (cf. [38]). Also, as illustrated in Table 6, the IDs sometimes contain the name of the speaker and/or the event during which the claim was uttered, both of which can be learned as a feature to match to verified target claims as well.

5.4 Comparison of individual datasets and splits

We next have a closer look at the training and test input and the set of verified claims for all datasets and compare characteristics such as claim length and word overlap. This provides insights into the task difficulty and the suitability of the data to assess the robustness and applicability of models to handle real-world claim retrieval scenarios.

5.4.1 Word overlap

In order to assess the difficulty of linking queries and target claims for the different datasets and the task in general, we measure the ratio of overlapping tokens/lemmas in input queries and targets, for both only the matching targets and all target verified claims including the false matches. For this, we tokenize and lemmatize the input using `spacy`³⁷, count the common tokens/lemmas for each pair of input query and target claim and normalize the count by the number of tokens in the smaller of the two sentences. This is because, as detailed in the task definition of 2020 (see Section 3), the target may be a sub-claim of the query or vice versa to justify a match and we do not want to give a higher weight to either of the directions in this analysis. We use the original data for the analysis, i.e. we do not perform de-duplication, as the original data is what has been used by the systems we strive to evaluate. As few systems use subtitle information, see Table 3, we omit them from this analysis.

Dataset	Split	Targets	$\cap_{\text{vclaim_T}}$	$\cap_{\text{vclaim_L}}$	$\cap_{\text{title_T}}$	$\cap_{\text{title_L}}$
2020-tweets	train	matches	0.345	0.396	0.252	0.336
2020-tweets	train	all	0.123	0.157	0.058	0.087
2020-tweets	test	matches	0.270	0.338	0.313	0.379
2020-tweets	test	all	0.073	0.100	0.041	0.062
2021-tweets	train	matches	0.341	0.393	0.250	0.334
2021-tweets	train	all	0.121	0.158	0.064	0.098
2021-tweets	test	matches	0.379	0.434	0.237	0.347
2021-tweets	test	all	0.132	0.171	0.067	0.103
2022-tweets	train	matches	0.342	0.393	0.250	0.334
2022-tweets	train	all	0.121	0.158	0.064	0.098
2022-tweets	test	matches	0.379	0.426	0.258	0.339
2022-tweets	test	all	0.134	0.169	0.0651	0.099
2021-debates	train	matches	0.385	0.452	0.197	0.247
2021-debates	train	all	0.156	0.213	0.064	0.098
2021-debates	test	matches	0.371	0.442	0.147	0.189
2021-debates	test	all	0.158	0.220	0.065	0.101
2022-debates	train	matches	0.385	0.452	0.197	0.247
2022-debates	train	all	0.156	0.213	0.064	0.098
2022-debates	test	matches	0.379	0.451	0.159	0.206
2022-debates	test	all	0.157	0.217	0.066	0.104

Table 7: Word (token (T) / lemma (L)) overlap between input queries and target verified claims (vclaim) and titles for the different datasets and splits relative to the number of tokens. Split: overlap between targets and queries from training or from test set. Targets: overlaps measured only for matching queries and targets or for all queries and all target verified claim pairs.

³⁷<https://spacy.io>

As Table 7 shows, the input queries and matching verified claims in the debates datasets have a higher token overlap than the ones in the tweets datasets (0.385 vs. 0.342 for train and 0.375 vs. 0.342 for test on average). For the non-matching pairs, overlaps are more frequent in the debates datasets as well (0.156 vs. 0.121 for train and 0.157 vs. 0.113 for test on average). Overlaps for matching queries and targets are generally high both in the training sets (mean: 0.359) and the test sets (mean: 0.355) for all datasets, while the overlaps between queries and all targets are much lower (train mean: 0.135, test mean: 0.130). The *2020-tweets* test set has a considerably lower overlap ratio between input queries and verified claims than all other datasets. While the overlaps are lower for the matching pairs as well, the difference between the overlap ratios of matching and all pairs is greater in this dataset, with the matching pairs having roughly 3.7 times as many overlaps compared to 2.5 (*2021-tweets*), 2.8 (*2022-tweets*), 2.3 (*2021-debates*) and 2.4 (*2022-debates*). A similar pattern can be observed for the titles. In contrast to the claim texts, the overlap ratios are higher in the tweets than in the debates datasets. For both discourse types, overlap ratios are higher for claim texts than for claim review titles.

While the overlap of lemmas naturally is higher than the overlap of tokens, the ratio of overlaps is already high without any pre-processing.

The performance of BM25-based baselines adds further insights regarding the similarities of input queries and matching verified claims for the different datasets. As these are included in the comparison of approaches in Section 6, we omit these statistics here.

5.4.2 Input query and target verified claim lengths

Figure 5 details the number of characters, respective tokens, in verified claims, verified claim review titles, and input claims.

Subfigures (a) and (d) of Figure 5 and Table 8 show that the distributions of verified claim lengths are similar in the different tweet datasets, with the mean of tokens ranging between 19.0 and 19.6, while debate claims are longer on average (22.7), but vary more strongly with a standard deviation of 9.6 vs. 7.4 - 7.5 for tweets. Verified claims matching input training and test queries are longer than the average of all verified claims for *2021-tweets* and *2022-tweets*, while the test input query matches are shorter for *2020-tweets* and the debates datasets. Even though *2021-debates* and *2022-debates* have different test queries, they are linked to the same targets (cf. Section 4) and thus share the same numbers.

The length of verified target claim titles (cf. Subfigures (c) and (f) of Figure 5 and Table 5) fluctuates more strongly across datasets (mean #tokens: 9.0 for *2020-tweets*, 10.4 for *2021-tweets* and *2022-tweets*, standard deviation 4.4 - 4.8 for *2020-tweets*) and they are longer in the debates data (mean #tokens: 13.6, standard deviation 4.0). Verified claim titles matching training queries are longer than all others for the tweets datasets and for *2021-tweets* and *2022-tweets*, the titles of claims matching test queries are longer than the average,

too, but not for *2020-tweets*. Titles contain overall fewer tokens than claim texts.

dataset	targets	vclaim_mean	vclaim_std	title_mean	title_std
2020-tweets	all	19.012	7.385	9.030	4.799
2020-tweets	train	21.046	6.958	13.192	3.698
2020-tweets	test	19.347	5.987	9.440	3.825
2021-tweets	all	19.501	7.353	10.361	4.370
2021-tweets	train	21.166	7.068	13.249	3.671
2021-tweets	test	24.357	9.585	12.469	2.918
2022-tweets	all	19.617	7.483	10.411	4.353
2022-tweets	train	21.160	7.081	13.249	3.674
2022-tweets	test	23.244	10.623	11.817	2.990
2021/2-debates	all	22.650	9.554	13.562	4.004
2021/2-debates	train	22.000	8.944	13.672	3.308
2021/2-debates	test	21.654	9.476	12.865	3.000

Table 8: Length of verified claims and their titles. Targets train/test: Claims c that match at least one input query in the training/test set

dataset	split	#tokens_mean	#tokens_std
2020-tweets	train	42.244	15.931
2020-tweets	test	29.755	5.815
2021-tweets	train	42.301	15.752
2021-tweets	test	44.540	16.765
2022-tweets	train	42.301	15.752
2022-tweets	test	45.459	16.645
2021-debates	train	24.107	16.342
2021-debates	test	22.684	15.008
2022-debates	train	24.107	16.342
2022-debates	test	24.292	15.407

Table 9: Input query lengths in training and test splits.

Similarly, the length of training input claims (cf. Subfigures (b) and (e) of Figure 5 and Table 9) is similar for the tweets datasets (mean #tokens: 42.3, standard deviation 15.8 - 15.9) with them being complete tweet texts with appended information. Debates training input claims are shorter (mean #tokens: 24.1, standard deviation 16.3), as they are individual statements uttered in a speech or debate.

A comparison of the last two columns of the figure and the corresponding Table 9 reveals that the distributions regarding input query length differs across training and test splits for all datasets except *2022-debates*, most strongly for the *2020-tweets* data. Striking is the difference between test and training input queries in *2020-tweets*, where the test queries are much shorter than the training queries (29.8 vs. 42.2) and have less variance regarding their length (5.8 vs. 15.9).

Many training input queries and target claims contain a very low or high number of tokens (cf. Subfigures (d) and (e) of Figure 5). Sentences with less than three tokens (excluding punctuation) can hardly constitute self-contained check-worthy or verifiable claims, as they should typically at least contain a subject, predicate, and object.

However, we count a small number of verified claims (29 for *2020-tweets*, 25 for *2021/2-tweets*), 9 for *2021/2-debates*) and 2 training input queries for the debates data with less than three non-punctuation tokens. None of these verified claims are correct matches for any training or test queries.

We examined the short queries and targets in more detail and found the latter to be due to extraction errors, e.g. as for the target claim from the 2021-tweets corpus in Figure 6 which consists merely of the word "The" or the target claim from *2022-tweets* in Figure 7 which has the verdict in the vclaim field. Such errors are hard to avoid when automatically extracting information from articles of fact-checking portals, as their structure is not fully consistent and the reviewed claims are not always explicitly listed, see, for instance, the fact-check at <https://www.snopes.com/fact-check/citibank/> which is part of *2020-tweets* where it is represented as

```
{"target_id": "4187", "target": "Citibank customers.",  
"title": "Citibank Phish"}.
```

The short input queries in the debates dataset are due to the fact that they may refer to the context, as outlined above and in Table 6, e.g. ID 20161019_3pres.0423: "Wrong", and ID 20160303_GOP_michigan.0575: "Ten times". This is in line with [3]'s analysis of [26]'s base debates dataset, where they found many claims that are not self-contained.

With this, they do not represent input that is expected in a real claim retrieval system, as users can typically be expected to enter a claim with its context. Note that, however, the context is accessible in the transcript files and systems participating in the Check That! Lab challenge may use them to enrich the input queries. When processed without additional context, debates input queries contain much less information (tokens and characters) than the tweets datasets.

As Figure 5 reveals, there is a long tail of very long claims and queries. We define outliers as those queries and claims whose token counts exceed the mean +2* standard deviation. The results of the analysis are listed in Table 10.

As Table 10 details, 3-4% of all target verified claims and those matching input training queries in the tweets datasets exceed the maximum expected length. The debates claims are not only longer on average, they also contain

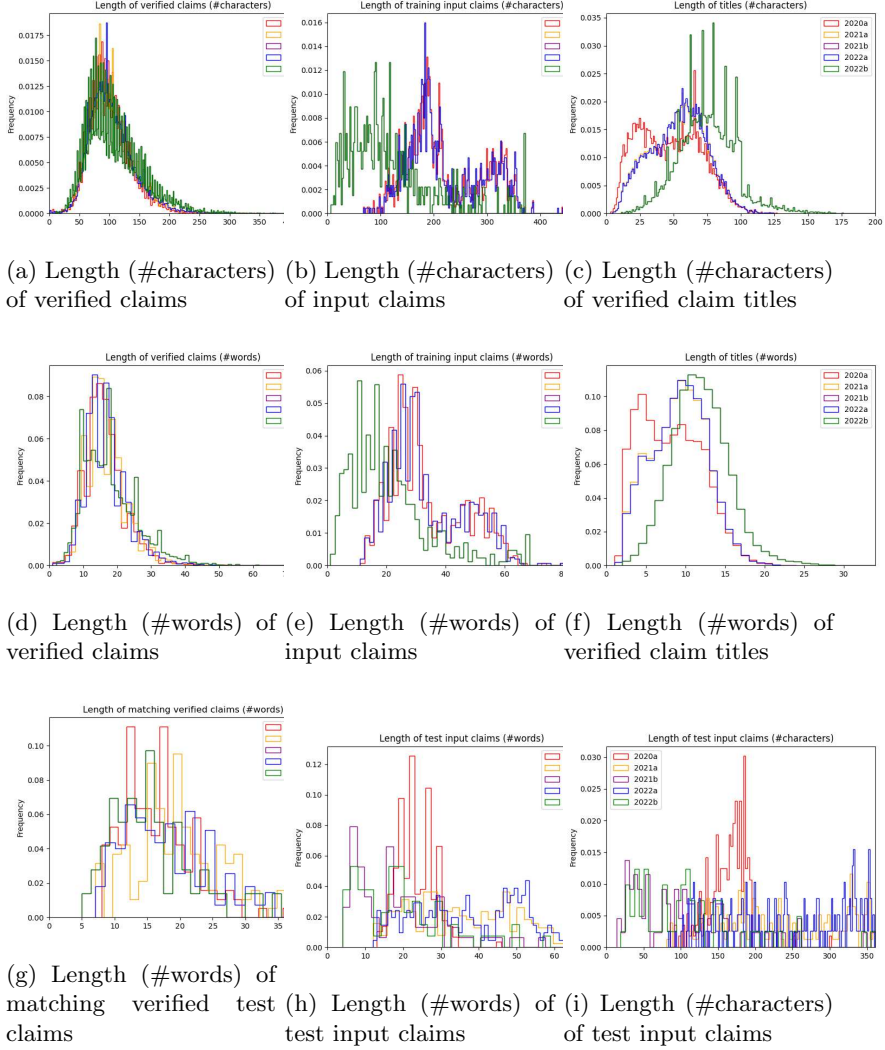


Fig. 5: Length of target verified claims, training and test input claims, and target verified claim titles in the different datasets.

more long outliers. For titles, we find between 2 and 3% to be outliers, and a proportionally slightly higher ratio of 4% for all test datasets except *2020-tweets*.

Outliers with fewer tokens than expected account for less than 1% in all datasets. We thus omit these numbers.

```
{
  "title": "3 Musketeers",
  "subtitle": "Rumor: The names of the Milky Way and 3
Musketeers candy bars were inadvertently switched.",
  "author": "David Mikkelson",
  "date": "13 June 2015",
  "vclaim_id": "vclaim-sno-3-musketeers",
  "vclaim": "The"
}
```

Fig. 6: Short vclaim in the set of 2021-tweets target verified claims due to an extraction error.

```
{
  "title": "Maryland School Bans Marine Veteran Over Daughter\u2019s
Homework",
  "subtitle": "Rumor claims a Marine was banned from the premises of
his daughter's school for objecting to a homework assignment about Islam.",
  "author": "David Mikkelson",
  "date": "2 November 2014",
  "vclaim_id": "vclaim-sno-marine-banned",
  "vclaim": "mixture"
}
```

Fig. 7: Short vclaim in the set of 2022-tweets target verified claims due to an extraction error.

Dataset	Targets	Threshold	Count
2020-tweets	all	33.8	328 (3.161%)
2020-tweets	train	34.9	25 (3.805%)
2020-tweets	test	31.3	8 (5.333%)
2021-tweets	all	34.2	486 (3.515%)
2021-tweets	train	35.3	25 (3.197%)
2021-tweets	test	43.5	5 (3.497%)
2022-tweets	all	34.6	543 (3.816%)
2022-tweets	train	35.3	25 (3.197%)
2022-tweets	test	44.5	10 (5.556%)
2021/2-debates	all	41.8	877 (4.556%)
2021/2-debates	train	39.9	19 (5.367%)
2021/2-debates	test	40.6	3 (5.769%)

Table 10: Long target verified claims classified as outliers regarding their numbers of tokens.

Dataset	Targets	Threshold	Count
2020-tweets	all	18.6	266 (2.564%)
2020-tweets	train	20.6	20 (3.044%)
2020-tweets	test	17.1	4 (2.667%)
2021-tweets	all	19.1	227 (1.642%)
2021-tweets	train	20.6	25 (3.197%)
2021-tweets	test	18.3	6 (4.196%)
2022-tweets	all	19.1	235 (1.651%)
2022-tweets	train	20.6	25 (3.197%)
2022-tweets	test	17.8	7 (3.889%)
2021/2-debates	all	21.6	626 (3.252%)
2021/2-debates	train	20.3	8 (2.260%)
2021/2-debates	test	18.9	2 (3.846%)

Table 11: Long target verified claim titles classified as outliers regarding their number of tokens.

5.5 Summary

The CheckThat! Lab claim retrieval datasets feature input queries and targets with many direct lexical overlaps. We can thus expect that claim retrieval methods relying on simple word overlap metrics can achieve high scores, even without lemmatization or similar pre-processing. Likewise, embedding-based systems can be expected to perform strongly when the vocabulary matches the dataset contents. Overlaps for queries with verified claim texts are higher than with titles and titles on average contain fewer tokens than the verified claims. Thus, titles might be less helpful when used in systems relying on lexical similarities to a large degree, but they could add more focused information with semantically similar concepts, as they usually are abstractive summaries of the claims.

Check-worthiness detection and claim retrieval are separate tasks. According to the task definition, input queries for the claim retrieval task are check-worthy claims or tweets. However, indeed input queries are not always individual, self-contained claims that constitute check-worthy claims in themselves as also visible in [26]’s analysis on their dataset that formed the basis of the Check That! Lab data. Therefore, models training on this data may not be able to distinguish essential parts of queries and targets, i.e. those that contain the check-worthy information, but only learn to find any matching information. Especially in light of the importance of word overlaps, the length of queries and verified claims can play a large role. Very short, long, or noisy claims can lead to overfitting. We also find erroneous claims in the datasets due to the difficulties of automatic harvesting from semi-structured fact-checking portals that can potentially increase the task difficulty. However, this mirrors data for real claim retrieval applications. Thus, investigating the robustness of methods regarding noise can yield valuable insights into the applicability of systems for applications.

The tweets datasets are similar to one another, and so are the debates datasets. *2020-tweets* stands out regarding word overlaps: here, word overlaps can be expected to be a better feature to distinguish matches from non-matches than for other datasets. This dataset also has shorter test queries with low variance regarding their length but high variance regarding the length of the training queries. This dataset contains less information for the verified claims, e.g. no date field. As dates are appended to the input tweets, models extracting this data for matching with the targets are expected to yield better results for 2021 and 2022 tweets data than for 2020.

In all datasets, the IDs give away some information: 1) their ordering (succeeding input queries may link to the same target) and 2) info about dates, events, speakers that can be extracted from them and used for linking (debates). When comparing the performance of different systems, it should be checked whether they make use of such information to gain a more realistic assessment of their performance in real-world systems.

In real-world applications, there may be no correct match for an input query while in the given data, there always is one match per input query for

the tweets datasets and at least one match per input query for the debates data. Nevertheless, the data can be used to develop methods that distinguish relevant from non-relevant claims. While the task is formulated as a ranking task, the gold data distinguishes relevant verified claims from irrelevant ones without ranking multiple valid matches.

6 Evaluation of Claim Retrieval Approaches

In this section, we present the evaluation of the claim retrieval systems following the experimental setup and research questions outlined above. In order to ensure comparability to the values specified in the literature, we use the original gold files including duplicates. However, we also measure the impact of the duplicates separately in order to reveal possible biases. While replicating the results for the systems, we found that the official results were generated using a fixed version of the *2020-tweets* gold file (cf. Section 5.1). We therefore use this fixed version for our comparative evaluation.

6.1 Performance on different datasets

Tables 19 and 20 detail the results of all approaches for training and testing on data of the same year and discourse type. Tables 14, 15, 16, 17 and 18 detail the performance by dataset. The MAP@k scores for *UofSheffield* remain constant for all values of k because this system returns only the top-1 matching verified claim per input query instead of a ranked list.

When only considering approaches that have participated in the challenge of a respective year (marked with $\star$), we can see that the top score was always achieved by a supervised method. Adding all approaches under investigation, we find that the unsupervised *SimBa 2023* outperforms the best supervised approach for *2020-tweets*, *BigIR* (cf. Table 14). The winner of the *2020-tweets* competition was *Buster.AI* with a score of 0.929. This approach ranks third in the overall comparison for this dataset. Surprising is the relatively weak performance of *RIET Lab*, which is the best-performing approach on all other datasets. This is due to one out of three runs where the model failed to learn maximally meaningful features: while the best run scored 0.957, the worst run achieved 0.801. For *2020-tweets*, but not the other datasets, two index-based approaches yield the top scores regarding precision (*BigIR* and *ES Baseline*), but not MAP, which includes recall. Also, the precision scores are generally higher than for the other datasets. The data analysis (Section 5) showed that *2020-tweets* features a relatively low token overlap of non-matching input queries and verified claims and a much higher overlap ratio in matching pairs. Thus, simple token matching techniques serve well to find matches with high precision. At the same time, absolute overlap ratios are lower than in the other tweet datasets, which hints at the presence of many pairs for which semantic similarity measures are required. In line with this, we find that models focusing strongly on embedding-based similarities, such as *SimBa 2023* and

Check_square: distmult, yield both higher precision and MAP scores on *2020-tweets* than on the other datasets even though the *ES Baseline* has its highest performance on this dataset, closely followed by *2022-tweets*.

For *2021-tweets*, the winning approach is *RIET Lab* (0.956), followed by *SimBa 2023* (0.945), *BirIR* (0.936), and the extension of the winner of the 2021 competition, *Aschern* (0.883): *CrowdChecked* (0.903). For the latest 2022 edition, the best performance on the tweets data is again achieved by *RIET Lab* with a score of 0.956 (cf. Table 16). *SimBa 2023* ranks second with 0.926, achieving comparable results to *AI Rational* (0.922) and *BigIR* (0.921). To the best of our knowledge, none of the approaches compared here uses information from the IDs as features, except *DIPS*, which extracts the dates from the input claims [38], and no approach exploits the order of input queries. Except for *SimBa 2023*, all approaches in the top 5 for the tweets data are supervised.

This is different for the debates data, where most unsupervised approaches outperform the supervised ones (cf. Tables 17 and 18), with the exception of *RIET Lab*, which is the best-performing approach for both 2021 and 2022. The unsupervised *SimBa 2023* outperforms the supervised *SimBa debates* system of 2022. In line with the observations from the data analysis (Section 5), claim retrieval is harder for the debates data than for the tweets data with the best-performing method achieving scores of ~ 0.5 while for the English tweets, the top scores reach ~ 0.96 MAP@5. The difficulty of the *2021-debates* and *2022-debates* test sets is similar, with the latter being easier with MAP@5 scores ranging from 0.238 - 0.501 vs. 0.196 - 0.480 for *2021-debates*.

The *ES Baseline* performs much lower than on the tweets data, yet the index-based approaches *ClaimLinker*, *TIET*, and *ES Baseline* work relatively well for the debates data compared to other systems. Their scores are generally similar to one another, except for *2020-tweets*. As detailed in Section 5, the difference between token/lemma overlap ratios in the negative vs. positive examples of the test set is the highest for this dataset. Thus, text overlap features seem to be particularly beneficial, which may explain *ClaimLinker*'s performance bonus compared to the *ES Baseline*.

Across datasets, the performance of the supervised approaches fluctuates considerably, even within the tweet datasets. Striking is the performance drop for *UofSheffield* for *2021-tweets*, and, to a lesser degree, *2022-tweets* (cf. Table 20), *NLP&IR@UNED*'s drop for *2021-tweets* as well as *RIET Lab*'s lower performance and high fluctuation on *2020-tweets*. The highest standard deviation for the supervised approaches is obtained by *NLP&IR@UNED* for *2022-tweets* with a maximum MAP@5 score of 0.506, a minimum of 0.235 and a standard deviation of 0.112 for all three runs.

Our data analysis in Section 5 revealed many outliers regarding verified claim length. Investigating the predictions in more detail, we find that *NLP&IR@UNED* has a strong bias of classifying very long claims (i.e. claims with more than 34 and 42 tokens, which are the average outlier ranges for the tweets and debates datasets, respectively, cf. Section 5.4.2) as correct matches for input queries (see Table 12). To generate these numbers, we counted for

each verified claim with excessive length how frequently their corresponding IDs occur at the top position and within the top-5 matches of the predictions files.

Dataset Top-k System	2020-tweets		2021-tweets		2022-tweets		2021-deb		2022-deb	
	1	5	1	5	1	5	1	5	1	5
Gold	7	7	25	25	24	24	5	5	2	2
ClaimLinker	5	10	22	33	20	32	6	17	4	11
TIET	7	27	26	107	30	109	5	37	3	28
SimBa 2023	7	21	24	77	26	98	6	28	3	23
UofSheffield	6	6	4	4	20	20	3	3	3	3
Check_square: distmult	3	29	25	111	27	127	8	26	5	21
Check_square run 1	6	32	15	63	19	77	5	27	6	17
Check_square run 2	3	32	21	70	18	71	6	24	7	21
Check_square run 3	3	29	18	77	17	74	6	21	6	19
NLP&IR@UNED run 1	15	176	79	382	150	719	5	17	14	87
NLP&IR@UNED run 2	8	69	90	539	174	746	8	73	6	39
NLP&IR@UNED run 3	31	341	89	381	192	804	4	40	8	56
RIET Lab run 1	6	30	25	85	25	92	10	27	6	21
RIET Lab run 2	8	30	26	106	26	96	9	26	5	19
RIET Lab run 3	3	26	25	87	25	94	12	29	6	20

Table 12: Number of verified claims with more than 34 (tweets) / 42 (debates) tokens in the systems’ top-5 predictions in C for the respective test sets.

Dataset Top-k System	2020-tweets		2021-tweets		2022-tweets	
	1	5	1	5	1	5
ClaimLinker	1	1	0	0	0	0
TIET	0	1	0	0	0	0
SimBa 2023	0	0	0	1	0	0
UofSheffield	0	0	7	7	2	2
Check_square: distmult	1	26	0	0	0	2
Check_square run 1	0	0	0	0	0	1
Check_square run 2	0	0	1	1	0	2
Check_square run 3	0	0	0	0	0	4
RIET Lab run 1	0	0	0	0	0	4
RIET Lab run 2	0	0	0	0	0	2
RIET Lab run 3	0	0	0	0	0	3

Table 13: Number of verified claims with less than 3 tokens (excluding punctuation) in the systems’ top-5 predictions in C for the respective test sets. Systems not including any of such claims are omitted in this table.

As revealed by Table 12, the counts vary heavily in between runs for *NLP&IR@UNED* and the counts are generally high. The other systems do not exhibit a similar bias, regardless of whether they use only vclaims or vclaims plus titles. All systems but *UofSheffield* rightfully include a smaller number of long claims in their predictions for *2020-tweets* than for the other tweet datasets. As outlined in Section 5, *2020-tweets* has considerably shorter test input queries with shorter matching verified claims, but the length of the matching training verified claims does not differ from the other tweets datasets. Note that *UofSheffield* only returns one claim per input query, and the gold files only list the top positions, thus their top-5 counts equal their top-1 counts. The erroneous and context-dependent short claims (cf. Section 5.4.2) also influence the results for some systems. As Table 13 shows, all supervised systems except *NLP&IR@UNED*, which is biased towards long claims, include some of the short claims in their predictions for *2020-tweets*. *UofSheffield* and *Check_square* additionally include them for *2021-tweets*. *UofSheffield*'s bias is particularly strong, as this system returns those claims at the top position in 9 cases. The unsupervised systems *ClaimLinker*, *TIET*, and *SimBa 2023* include only one short claim in total each, for either *2020-tweets* or *2021-tweets*. The unsupervised *Check_square: distmult* system returns a high number of short claims in its top-5 predictions for *2020-tweets* and two of them for *2022-tweets*. The difference to the supervised variant is the choice of embeddings: fine-tuned Sentence-BERT embeddings in *Check_square*, pre-trained multilingual DistilBERT embeddings in *Check_square: distmult*. While it overall achieves higher performance than the variant using fine-tuned BERT embeddings, the fine-tuning using triplets that include hard negatives with high cosine similarity seems to be effective in increasing robustness to this kind of noise.

This analysis supports the interpretation that the supervised approaches may learn spurious correlations in this data and may thus be sensitive to noise and less stable regarding their performance when applied to real data in claim retrieval systems than unsupervised approaches. At the same time, lower performance on the given data does not necessarily mean lower task performance. Instead, a system may be worse at picking up spurious correlations, as explained by [8]. We find hints for this in our comparison of *Check_square* using fine-tuned BERT vs. pre-trained DistilBERT embeddings and the greater preference of the latter for choosing long claims as matches for *2020-tweets*. A related observation was made by [39] who found that fine-tuning using titles did not improve their system, but improved its robustness. In general, using the more abstractive titles can intuitively be expected to help a system deal with paraphrased input claims, and many systems use them in addition to the vclaim texts, cf. Table 3. However, the methodology of dataset collection might have introduced biases, e.g. fact-checkers listing tweets or statements they checked in their articles might be likely to phrase their summarizing claims in a similar way. In this case, the importance of abstractive titles as an additional information source might be pronounced in real-world data. However, the correlations present in the data may also reflect properties of real-world data, i.e.

input queries and matching verified claims naturally share many tokens, such as entities mentioned in their most common form.

As the different behavior of the systems is not always visible using all of the datasets, this analysis highlights that testing on the different dataset is important to assess model performance and biases for this task.

Our results generally verify the observations made in the literature that the choice of language models for this task greatly impacts the results: For *Check_square*, the authors found on the *2021-tweets* data that multilingual DistilBERT embeddings without fine-tuning outperformed fine-tuned monolingual Sentence-BERT models. For the former, fine-tuning using their triplet loss methodology hurts performance while it helps in the case of Sentence-BERT [45]. Similarly, [39] found on the same dataset that their RoBERTa model fine-tuned on triplets performed worse than monolingual DistilBERT without fine-tuning. We could replicate that *Check_square*'s variant using multilingual DistilBERT embeddings without fine-tuning outperforms the fine-tuned BERT embeddings on all datasets but *2022-tweets*. For this dataset, both variants perform on par. Similarly, according to [26], Sentence-BERT and Universal Sentence Encoder performed much better in their system and on their data than BERT, and Sentence-BERT better than Universal Sentence Encoder. In the same vain, [34] found on *2020-tweets* that using a standard BERT model to represent sentences resulted in low performance while pre-trained Sentence-BERT performed a lot higher and further benefited from fine-tuning. [31] found that BERT performs poorly in comparison with other models such as RoBERTa, MiniLM, or MPNet in their *BigIR* system, which they tested for *2020-tweets* and *2021-tweets*. They find MPNet to yield the best performance. MPNet also yielded the best performance in [47]'s analysis on the *2022-tweets* data, followed by SimCSE-RoBERTA-large and Universal Sentence Encoder. InferSent's performance turned out to be considerably worse, both using fast-Text and GloVe. The two dominating systems, *RIET Lab* and *SimBa 2023*, both use sentence transformers for blocking: fine-tuned Sentence-T5-Large for *RIET Lab*, pre-trained MPNet-Base for *SimBa 2023*. Sentence-T5 is also used for re-ranking by *SimBa 2023* in its smaller pre-trained base variant. *Check_square* and *TIET* rely on standard Sentence-BERT embeddings. Their relatively weak performance is in line with these findings and can thus most likely be attributed to the language model choice. Consistent with this, their performance on *2021-tweets* is lower than the baseline RoBERTa performance shown by *NLytics*. Also, while *CrowdChecked* benefits from the additional training data (cf. [30]), it relies on Sentence-BERT and is outperformed by approaches using better-performing language models. This supports the finding that the language model is the most important factor in solving this task, more important than acquiring additional training data and developing sophisticated training/fine-tuning strategies. The latter finding is also supported by the comparison of *SimBa 2023* and *RIET Lab*, whose performance is close, even though *RIET Lab* applies considerable training and fine-tuning effort on

Rank	Sup.?	Method	P@1	P@3	P@5	MAP@1	MAP@3	MAP@5
1		SimBa 2023	0.945	0.328	0.197	0.945	0.964	0.964
2	✓	BigIR*	0.950	*	*	*	*	0.955
3	✓	Buster.AI**	0.895	0.320	0.195	0.897	0.926	0.929
4	✓	UNIPI-NLE**	0.875	0.315	0.193	0.877	0.907	0.912
5	✓	UB_ET**	0.815	0.307	0.186	0.818	0.862	0.864
6	✓	RIET-Lab	0.827	0.297	0.179	0.827	0.856	0.857
7	✓	NLP&IR@UNED**	0.805	0.300	0.185	0.807	0.851	0.856
8		NLP&IR@UNED	0.798	0.294	0.179	0.798	0.837	0.840
9		Check_square: distmult	0.795	0.292	0.181	0.795	0.829	0.836
10		ClaimLinker	0.810	0.283	0.170	0.810	0.828	0.828
11	✓	UofSheffield	0.815	0.272	0.163	0.815	0.815	0.815
12	✓	UofSheffield**	0.805	0.270	0.162	0.807	0.807	0.807
13	✓	Check_square	0.758	0.280	0.173	0.758	0.797	0.802
14		TIET	0.745	0.278	0.180	0.745	0.785	0.800
15		ES Baseline	0.793	0.344	0.217	0.682	0.782	0.794
16		TIET**	0.740	0.267	0.164	0.743	0.768	0.773
17	✓	Check_square**	0.650	0.247	0.152	0.652	0.690	0.695
18		ES Baseline 2020**	0.472	0.249	0.156	0.470	0.601	0.609

Table 14: Results for *2020-tweets*. Sup.? = Supervised/fine-tuned system? Numbers for approaches marked with * have been taken from the respective publications. Scores not detailed therein are marked with * in the respective columns. All other numbers we generated/replicated for this study. Systems marked with * participated in the challenge of that year

very large language models while *SimBa 2023* employs high-performing language models in their pre-trained base variants without any supervision. For *SimBa 2023* and its choice of language models combined with lexical features, we find no disadvantage regarding model robustness.

Finally, we repeated the analysis using the de-duplicated gold files. While the scores deviate for datasets with duplicates, *2021-debates* and *2022-debates*, cf. Section 5, e.g. on *2022-debates* 0.478 on the de-duplicated gold data vs. 0.501 for *RIET Lab* and 0.461 on the de-duplicated gold data vs. 0.481 for *SimBa 2023*, the order of the models regarding their performance and the differences between the datasets remain unchanged. We therefore omit details of these scores.

Rank	Sup.?	Method	P@1	P@3	P@5	MAP@1	MAP@3	MAP@5
1	✓	RIET-Lab	0.926	0.329	0.199	0.926	0.955	0.956
2		SimBa 2023	0.921	0.322	0.195	0.921	0.943	0.945
3	✓	BigIR*	0.911	*	*	*	*	0.936
4	✓	CrowdChecked*	*	*	*	*	*	0.903
5	✓	Aschern**	0.861	0.300	0.182	0.861	0.880	0.883
6		NLytics**	0.738	0.289	0.179	0.738	0.792	0.799
7	✓	DIPS*	0.728	0.282	0.177	0.728	0.778	0.787
8		TIET	0.683	0.271	0.173	0.683	0.741	0.753
9		ES Baseline	0.703	0.262	0.164	0.703	0.741	0.749
9		ES Baseline 2021*	0.703	0.262	0.164	0.703	0.741	0.749
10		ClaimLinker	0.708	0.256	0.159	0.708	0.735	0.742
11		Check_square: distmult	0.673	0.257	0.167	0.673	0.717	0.732
12	✓	Check_square	0.630	0.251	0.160	0.630	0.684	0.695
13	✓	NLP&IR@UNED	0.470	0.234	0.156	0.470	0.575	0.592
14	✓	UofSheffield	0.292	0.097	0.058	0.292	0.292	0.292

Table 15: Results for *2021-tweets*. Sup.? = Supervised/fine-tuned system? Numbers for approaches marked with * have been taken from the respective publications. Scores not detailed therein are marked with * in the respective columns. All other numbers we generated/replicated for this study. Systems marked with ★ participated in the challenge of that year

Rank	Sup.?	Method	P@1	P@3	P@5	MAP@1	MAP@3	MAP@5
1	✓	RIET-Lab**	*	0.322	0.194	0.943	0.955	0.956
2	✓	RIET-Lab	0.928	0.324	0.194	0.928	0.948	0.948
3		SimBa 2023	0.904	0.319	0.191	0.904	0.926	0.926
4	✓	AI Rational*	*	0.313	0.190	0.904	0.919	0.922
5	✓	BigIR**	*	0.316	0.189	0.900	0.921	0.921
6		SimBa (tweets)**	*	0.314	0.190	0.876	0.905	0.907
7	✓	Fraunhofer SIT SVM**	0.752	0.275	0.170	0.752	0.787	0.794
8		ES Baseline	0.756	0.276	0.168	0.756	0.789	0.791
9		ClaimLinker	0.751	0.265	0.161	0.751	0.771	0.773
10		TIET	0.708	0.274	0.174	0.708	0.760	0.771
11	✓	Check_square	0.705	0.266	0.167	0.705	0.745	0.754
12		Check_square: distmult	0.703	0.266	0.166	0.703	0.746	0.752
13	✓	UofSheffield	0.679	0.226	0.136	0.679	0.679	0.679
14	✓	Fraunhofer SIT**	0.557	0.221	0.141	0.557	0.601	0.610
15	✓	NLP&IR@UNED	0.215	0.184	0.134	0.215	0.354	0.381

Table 16: Results for *2022-tweets*. Sup.? = Supervised/fine-tuned system? Numbers for approaches marked with * have been taken from the respective publications. Scores not detailed therein are marked with * in the respective columns. All other numbers we generated/replicated for this study. Systems marked with ★ participated in the challenge of that year

Rank	Sup.?	Method	P@1	P@3	P@5	MAP@1	MAP@3	MAP@5
1	✓	RIET-Lab	0.414	0.205	0.127	0.414	0.475	0.480
2		SimBa 2023	0.405	0.177	0.109	0.411	0.439	0.441
3		ClaimLinker	0.346	0.158	0.095	0.353	0.385	0.385
4		TIET	0.316	0.148	0.091	0.316	0.351	0.353
5		ES Baseline	0.304	0.144	0.091	0.304	0.339	0.346
5		ES Baseline 2021*	0.304	0.143	0.091	0.304	0.339	0.346
6	✓	DIPS*★	0.266	0.143	0.099	0.278	0.313	0.328
7	✓	Beasku*★	0.253	0.139	0.101	0.266	0.308	0.327
8		Check_square: distmult	0.266	0.131	0.089	0.259	0.295	0.304
9	✓	Check_square	0.236	0.105	0.074	0.224	0.247	0.259
10	✓	NLP&IR@UNED	0.181	0.101	0.066	0.171	0.216	0.221
11		NLytics*★	0.165	0.101	0.068	0.171	0.210	0.215
12	✓	UofSheffield	0.203	0.072	0.043	0.196	0.196	0.196

Table 17: Results for *2021-debates*. Sup.? = Supervised/fine-tuned system? Numbers for approaches marked with * have been taken from the respective publications. Scores not detailed therein are marked with * in the respective columns. All other numbers we generated/replicated for this study. Systems marked with ★ participated in the challenge of that year

Rank	Sup.?	Method	P@1	P@3	P@5	MAP@1	MAP@3	MAP@5
1	✓	RIET-Lab	0.441	0.222	0.137	0.415	0.497	0.501
2		SimBa 2023	0.462	0.190	0.120	0.446	0.474	0.481
3	✓	SimBa (debates)*★	*	0.190	0.126	0.408	0.446	0.459
4		ClaimLinker	0.385	0.169	0.105	0.369	0.408	0.411
5		TIET	0.338	0.154	0.095	0.323	0.365	0.367
6		ES Baseline	0.323	0.154	0.099	0.308	0.353	0.361
7		Check_square: distmult	0.323	0.149	0.098	0.300	0.336	0.347
8	✓	Check_square	0.277	0.135	0.084	0.249	0.298	0.300
9	✓	NLP&IR@UNED	0.226	0.113	0.082	0.215	0.249	0.261
10	✓	UofSheffield	0.246	0.087	0.052	0.238	0.238	0.238

Table 18: Results for *2022-debates*. Sup.? = Supervised/fine-tuned system? Numbers for approaches marked with * have been taken from the respective publications. Scores not detailed therein are marked with * in the respective columns. All other numbers we generated/replicated for this study. Systems marked with ★ participated in the challenge of that year

Method	Dataset	P@1	P@3	P@5	MAP@1	MAP@3	MAP@5
Check_square: distmult	<u>2020-tweets</u>	0.790	0.292	0.181	0.792	0.827	0.834
	2021-tweets	0.673	0.257	0.167	0.673	0.717	0.732
	2022-tweets	0.703	0.266	0.166	0.703	0.746	0.752
	2021-debates	0.266	0.131	0.089	0.259	0.295	0.304
	2022-debates	0.323	0.149	0.098	0.300	0.336	0.347
SimBa 2023	<u>2020-tweets</u>	0.940	0.328	0.197	0.943	0.962	0.962
	2021-tweets	0.921	0.322	0.195	0.921	0.943	0.945
	2022-tweets	0.904	0.319	0.191	0.904	0.926	0.926
	2021-debates	0.405	0.177	0.109	0.411	0.439	0.441
	2022-debates	0.462	0.190	0.120	0.446	0.474	0.481
SimBa (tweets)*	2022-tweets*	*	0.314	0.190	0.876	0.905	0.907
ClaimLinker	<u>2020-tweets</u>	0.800	0.282	0.170	0.800	0.820	0.820
	2021-tweets	0.693	0.256	0.160	0.693	0.727	0.735
	2022-tweets	0.742	0.265	0.161	0.742	0.766	0.768
	2021-debates	0.333	0.124	0.074	0.333	0.346	0.346
	2022-debates	0.323	0.133	0.083	0.323	0.362	0.365
TIET	<u>2020-tweets</u>	0.745	0.278	0.180	0.745	0.785	0.800
	2021-tweets	0.683	0.271	0.173	0.683	0.741	0.753
	2022-tweets	0.708	0.274	0.174	0.708	0.760	0.771
	2021-debates	0.316	0.148	0.091	0.316	0.351	0.353
	2022-debates	0.338	0.154	0.095	0.323	0.365	0.367
NLytics*	<u>2021-tweets*</u>	0.738	0.289	0.179	0.738	0.792	0.799
	2021-debates	0.165	0.101	0.068	0.171	0.210	0.215
ES Baseline	<u>2020-tweets</u>	0.793	0.344	0.217	0.682	0.782	0.794
	2021-tweets*	0.703	0.262	0.164	0.703	0.741	0.749
	2021-tweets	0.703	0.262	0.164	0.703	0.741	0.749
	2022-tweets	0.756	0.276	0.168	0.756	0.789	0.791
	2021-debates*	0.304	0.143	0.091	0.304	0.339	0.346
	2021-debates	0.304	0.144	0.091	0.304	0.339	0.346
	2022-debates	0.323	0.154	0.099	0.308	0.353	0.361
ES Baseline 2020*	2020-tweets	0.472	0.249	0.156	0.470	0.601	0.609

Table 19: Performance of the unsupervised approaches on the different datasets. Numbers for approaches marked with * have been taken from the respective publications. Scores not detailed therein are marked with * in the respective columns. All other numbers we generated/replicated for this study. Boldface marks the best scores on a dataset for the supervised approaches, underlined scores are the best scores for all approaches, including the supervised ones (cf. Table 20). Blue color marks the best scores of an approach across datasets.

Method	Dataset	P@1	P@3	P@5	MAP@1	MAP@3	MAP@5
Check_square	2020-tweets*	0.650	0.247	0.152	0.652	0.690	0.695
	2020-tweets	0.753	0.280	0.173	0.756	0.794	0.799
	2021-tweets	0.630	0.251	0.160	0.630	0.684	0.695
	2022-tweets	0.705	0.266	0.167	0.705	0.745	0.754
	2021-debates	0.236	0.105	0.074	0.224	0.247	0.259
	2022-debates	0.277	0.135	0.084	0.249	0.298	0.300
RIET-Lab	2020-tweets	0.822	0.297	0.179	0.824	0.853	0.854
	2021-tweets	0.926	0.329	0.199	0.926	0.955	0.956
	2022-tweets*	*	0.322	<u>0.194</u>	<u>0.943</u>	<u>0.955</u>	<u>0.956</u>
	2022-tweets	0.928	0.324	0.194	0.928	0.948	0.948
	2021-debates	0.414	0.205	0.127	0.414	0.475	0.480
	2022-debates	0.441	0.222	0.137	0.415	0.497	0.501
UofSheffield	2020-tweets*	0.805	0.270	0.162	0.807	0.807	0.807
	2020-tweets	0.810	0.272	0.163	0.812	0.812	0.812
	2021-tweets	0.292	0.097	0.058	0.292	0.292	0.292
	2022-tweets	0.679	0.226	0.136	0.679	0.679	0.679
	2021-debates	0.203	0.072	0.043	0.196	0.196	0.196
	2022-debates	0.246	0.087	0.052	0.238	0.238	0.238
NLP&IR@UNED	2020-tweets*	0.805	0.300	0.185	0.807	0.851	0.856
	2020-tweets	0.795	0.294	0.179	0.797	0.835	0.838
	2021-tweets	0.470	0.234	0.156	0.470	0.575	0.592
	2022-tweets	0.215	0.184	0.134	0.215	0.354	0.381
	2021-debates	0.181	0.101	0.066	0.171	0.216	0.221
	2022-debates	0.226	0.113	0.082	0.215	0.249	0.261
BigIR*	2020-tweets*	0.950	*	*	*	*	0.955
	2021-tweets*	0.911	*	*	*	*	0.936
	2022-tweets*	*	0.316	0.189	0.900	0.921	0.921
SimBa (debates)*	2022-debates*	*	0.190	0.126	0.408	0.446	0.459
AI Rational*	2022-tweets*	*	0.313	0.190	0.904	0.919	0.922
Buster.AI*	2020-tweets*	0.895	0.320	0.195	0.897	0.926	0.929
UNIPi-NLE*	2020-tweets*	0.875	0.315	0.193	0.877	0.907	0.912
UB.ET*	2020-tweets*	0.815	0.307	0.186	0.818	0.862	0.864
CrowdChecked*	2021-tweets	*	*	*	*	*	0.903
Aschern*	2021-tweets*	0.861	0.300	0.182	0.861	0.880	0.883
DIPS*	2021-tweets*	0.728	0.282	0.177	0.728	0.778	0.787
	2021-debates*	0.266	0.143	0.099	0.278	0.313	0.328
Beasku*	2021-debates*	0.253	0.139	0.101	0.266	0.308	0.327
Fraunhofer SIT*	2022-tweets*	0.557	0.221	0.141	0.557	0.601	0.610
Fraunhofer SIT SVM*	2022-tweets*	0.752	0.275	0.170	0.752	0.787	0.794

Table 20: Performance of the supervised approaches on the different datasets. Numbers for approaches marked with * have been taken from the respective publications. Scores not detailed therein are marked with * in the respective columns. All other numbers we generated/replicated for this study. Boldface marks the best scores on a dataset for the supervised approaches, underlined scores are the best scores for all approaches, including the unsupervised ones (cf. Table 19). Blue color marks the best scores of an approach across datasets.

6.2 Cross-dataset evaluation

The cross-evaluation results are displayed in Tables 21 and 22. Since we reported averaged results in the previous subsection, the performance of a trained model on the test dataset of the same type and year does not correspond to the values in Table 20. We chose a random run (e.g. always the model trained in the last run) rather than selecting a specific model. As the results show, training an approach on a dataset of a different year or type does not necessarily hurt performance. Instead, we find that models that perform well on one dataset perform well on others, too. E.g. NLP&IR@UNED and Uof-Sheffield learnt their most useful model on *2020-tweets*. This yields the best results on all datasets. The same can be observed for RIET Lab with its models for *2021-tweets* and *2022-tweets*. Check_square learnt its least useful model on *2021-debates* which scores worse than all models learnt on any other dataset for all test data, including *2022-debates* which contains data of the same discourse type. Together with the insights from the previous subsection which showed that individual training runs led to great variation in performance, this finding highlights that the supervised approaches are vulnerable to overfitting, making their performance hard to predict. However, when a model is learnt successfully for this task, it generalizes well across data of different discourse types. Training on *2020-tweets* generally yields the best results, with the exception of RIET Lab and Check_square, for which the model trained on *2022-tweets* achieves better scores on the *2020-tweets* test set and the models learnt on *2021-tweets* and on *2020-tweets* perform on par on the *2021-debates* test set. As outlined in Section 5, tweet datasets have a higher number of positive training examples than the debates datasets and a smaller number of verified claims to match against. At the same time, retrieval of matching pairs depends largely on token overlaps for both the tweets and the debates data. These characteristics explain why models learnt on the tweets data perform better than models learnt on the debates datasets, even when applied to data of the other discourse type. *2020-tweets* has a smaller training set than the other tweet datasets, but also a smaller number of non-matching verified claims, i.e. negative training examples, and the ratio of positive vs. negative examples has a lesser bias towards negative examples. Unless a model overfits the data, training on *2020-tweets* thus yields the best models.

To control for biases introduced by the duplicates in the gold files, we repeated the evaluation with the de-duplicated versions. As in the dataset-wise comparison in the previous subsection, we found no biases and omit the scores.

6.3 Efficiency

The systems we compare exhibit the following runtime complexities for the online computing steps, with D denoting the size of the verified claim database, i.e. the number of verified candidate claims to match against, and $C\{n\}$ unknown constants, see Table 23.

Method	Dataset (training → test)	P@1	P@3	P@5	MAP@1	MAP@3	MAP@5
Check_square	2020-tweets → 2020-tweets	0.755	0.277	0.170	0.757	0.789	0.794
	2021-tweets → 2020-tweets	0.735	0.272	0.170	0.738	0.772	0.780
	<u>2022-tweets → 2020-tweets</u>	<u>0.795</u>	<u>0.290</u>	<u>0.177</u>	<u>0.797</u>	<u>0.829</u>	<u>0.832</u>
	2021-debates → 2020-tweets	0.640	0.245	0.155	0.642	0.683	0.692
	<u>2020-tweets → 2021-tweets</u>	<u>0.733</u>	<u>0.274</u>	<u>0.172</u>	<u>0.733</u>	<u>0.771</u>	<u>0.781</u>
	2021-tweets → 2021-tweets	0.619	0.231	0.151	0.619	0.653	0.667
	2022-tweets → 2021-tweets	0.663	0.248	0.157	0.663	0.701	0.712
	2021-debates → 2021-tweets	0.475	0.203	0.131	0.475	0.538	0.548
	<u>2020-tweets → 2022-tweets</u>	<u>0.727</u>	0.274	0.168	<u>0.727</u>	<u>0.770</u>	<u>0.775</u>
	2021-tweets → 2022-tweets	0.679	0.255	0.157	0.679	0.718	0.722
	2022-tweets → 2022-tweets	0.713	<u>0.276</u>	<u>0.171</u>	0.713	0.760	0.766
	2021-debates → 2022-tweets	0.560	0.220	0.138	0.560	0.605	0.612
	2020-tweets → 2021-debates	<u>0.253</u>	<u>0.143</u>	<u>0.091</u>	<u>0.247</u>	<u>0.304</u>	0.309
	<u>2021-tweets → 2021-debates</u>	<u>0.253</u>	<u>0.143</u>	<u>0.091</u>	<u>0.247</u>	<u>0.304</u>	<u>0.310</u>
	2022-tweets → 2021-debates	0.241	0.135	0.089	0.247	0.293	0.302
	2021-debates → 2021-debates	0.228	0.110	0.073	0.215	0.249	0.258
	<u>2020-tweets → 2022-debates</u>	<u>0.292</u>	<u>0.159</u>	0.102	<u>0.277</u>	<u>0.337</u>	<u>0.343</u>
	2021-tweets → 2022-debates	<u>0.292</u>	0.154	0.102	<u>0.277</u>	0.331	0.342
	2022-tweets → 2022-debates	0.277	0.149	<u>0.105</u>	<u>0.277</u>	0.322	0.340
	2021-debates → 2022-debates	0.262	0.128	0.086	0.231	0.279	0.291
RIET-Lab	2020-tweets → 2020-tweets	0.765	0.285	0.171	0.767	0.805	0.805
	2021-tweets → 2020-tweets	0.940	<u>0.332</u>	<u>0.199</u>	0.943	<u>0.967</u>	0.967
	<u>2022-tweets → 2020-tweets</u>	<u>0.945</u>	0.330	<u>0.199</u>	<u>0.948</u>	<u>0.967</u>	<u>0.968</u>
	2021-debates → 2020-tweets	0.910	<u>0.332</u>	<u>0.199</u>	0.912	0.950	0.950
	2020-tweets → 2021-tweets	0.881	0.317	0.195	0.881	0.913	0.919
	<u>2021-tweets → 2021-tweets</u>	<u>0.926</u>	0.330	<u>0.199</u>	<u>0.926</u>	0.956	<u>0.958</u>
	<u>2022-tweets → 2021-tweets</u>	<u>0.926</u>	<u>0.332</u>	<u>0.199</u>	<u>0.926</u>	<u>0.958</u>	<u>0.958</u>
	2021-debates → 2021-tweets	0.871	0.315	0.190	0.871	0.908	0.909
	2020-tweets → 2022-tweets	0.919	0.322	<u>0.195</u>	0.919	0.942	0.944
	<u>2021-tweets → 2022-tweets</u>	<u>0.933</u>	<u>0.324</u>	0.194	<u>0.933</u>	<u>0.951</u>	<u>0.951</u>
	2022-tweets → 2022-tweets	0.928	<u>0.324</u>	0.194	0.928	0.946	0.946
	2021-debates → 2022-tweets	0.866	0.308	0.186	0.866	0.892	0.893
	2020-tweets → 2021-debates	0.418	0.181	0.119	0.411	0.444	0.454
	<u>2021-tweets → 2021-debates</u>	<u>0.481</u>	<u>0.207</u>	0.124	<u>0.481</u>	<u>0.514</u>	<u>0.514</u>
	2022-tweets → 2021-debates	0.468	<u>0.207</u>	0.127	0.468	0.507	0.510
	2021-debates → 2021-debates	0.418	<u>0.207</u>	<u>0.129</u>	0.418	0.482	0.488
	2020-tweets → 2022-debates	0.492	0.205	0.135	0.462	0.499	0.511
	<u>2021-tweets → 2022-debates</u>	<u>0.554</u>	<u>0.231</u>	0.138	<u>0.531</u>	<u>0.573</u>	<u>0.573</u>
	2022-tweets → 2022-debates	0.538	<u>0.231</u>	<u>0.142</u>	0.515	0.563	0.566
	2021-debates → 2022-debates	0.477	<u>0.231</u>	<u>0.142</u>	0.454	0.529	0.533

Table 21: Cross-evaluation results for supervision and fine-tuning. Boldface marks the best score on a test dataset across all approaches (including those listed in Table 22), underline the best score across all datasets and approaches, blue color the best scores for an approach for each test dataset.

Note that using the Sentence Transformer, embeddings are projected to Euclidean space which enables search with $O(i+D)$ inferences with i being the number of input claims, instead of $O(i * D)$ pair-wise comparisons (cf. [46]).

For real applications, especially web-based ones, the magnitudes of the constants matter.

As detailed in Subsections 6.1 and 6.2, RIET Lab is the approach with the highest performance on most datasets but has very high online and offline

Method	Dataset (training → test)	P@1	P@3	P@5	MAP@1	MAP@3	MAP@5
UofSheffield	2020-tweets → 2020-tweets	0.810	0.272	0.163	0.812	0.812	0.812
	2021-tweets → 2020-tweets	0.640	0.215	0.129	0.642	0.642	0.642
	2022-tweets → 2020-tweets	0.775	0.260	0.156	0.777	0.777	0.777
	2021-debates → 2020-tweets	0.795	0.267	0.160	0.797	0.797	0.797
	2020-tweets → 2021-tweets	0.752	0.251	0.150	0.752	0.752	0.752
	2021-tweets → 2021-tweets	0.292	0.097	0.058	0.292	0.292	0.292
	2022-tweets → 2021-tweets	0.698	0.233	0.140	0.698	0.698	0.698
	2021-debates → 2021-tweets	0.723	0.241	0.145	0.723	0.723	0.723
	2020-tweets → 2022-tweets	0.742	0.247	0.148	0.742	0.742	0.742
	2021-tweets → 2022-tweets	0.301	0.100	0.060	0.301	0.301	0.301
	2022-tweets → 2022-tweets	0.679	0.226	0.136	0.679	0.679	0.679
	2021-debates → 2022-tweets	0.689	0.230	0.138	0.689	0.689	0.689
	2020-tweets → 2021-debates	0.304	0.118	0.071	0.310	0.310	0.310
	2021-tweets → 2021-debates	0.228	0.084	0.051	0.228	0.228	0.228
	2022-tweets → 2021-debates	0.278	0.105	0.063	0.278	0.278	0.278
	2021-debates → 2021-debates	0.203	0.072	0.043	0.196	0.196	0.196
	2020-tweets → 2022-debates	0.338	0.123	0.074	0.331	0.331	0.331
	2021-tweets → 2022-debates	0.262	0.092	0.055	0.254	0.254	0.254
	<u>2022-tweets → 2022-debates</u>	<u>0.338</u>	<u>0.123</u>	<u>0.074</u>	0.323	0.323	0.323
	2021-debates → 2022-debates	0.246	0.087	0.052	0.238	0.238	0.238
NLP&IR@UNED	2020-tweets → 2020-tweets	0.805	0.297	0.180	0.807	0.843	0.846
	2021-tweets → 2020-tweets	0.445	0.187	0.152	0.445	0.497	0.543
	2022-tweets → 2020-tweets	0.070	0.063	0.107	0.070	0.118	0.199
	2021-debates → 2020-tweets	0.690	0.277	0.171	0.690	0.752	0.758
	2020-tweets → 2021-tweets	0.713	0.285	0.180	0.713	0.776	0.787
	2021-tweets → 2021-tweets	0.475	0.234	0.160	0.475	0.579	0.602
	2022-tweets → 2021-tweets	0.307	0.205	0.145	0.307	0.446	0.471
	2021-debates → 2021-tweets	0.569	0.219	0.137	0.569	0.611	0.617
	2020-tweets → 2022-tweets	0.842	0.292	0.179	0.842	0.858	0.862
	2021-tweets → 2022-tweets	0.584	0.273	0.171	0.584	0.691	0.700
	2022-tweets → 2022-tweets	0.344	0.223	0.152	0.344	0.485	0.506
	2021-debates → 2022-tweets	0.612	0.220	0.134	0.612	0.635	0.637
	2020-tweets → 2021-debates	0.253	0.122	0.086	0.234	0.272	0.283
	2021-tweets → 2021-debates	0.215	0.089	0.061	0.196	0.213	0.219
	2022-tweets → 2021-debates	0.152	0.068	0.043	0.146	0.156	0.159
	2021-debates → 2021-debates	0.215	0.105	0.071	0.203	0.241	0.248
	2020-tweets → 2022-debates	0.308	0.149	0.105	0.269	0.323	0.337
	2021-tweets → 2022-debates	0.277	0.118	0.080	0.246	0.274	0.282
	2022-tweets → 2022-debates	0.185	0.082	0.055	0.177	0.190	0.195
	2021-debates → 2022-debates	0.262	0.123	0.083	0.231	0.274	0.284

Table 22: Cross-evaluation results for supervision and fine-tuning. Boldface marks the best score on a test dataset across all approaches (including those listed in Table 21), underline the best score across all datasets and approaches, blue color the best scores for an approach for each test dataset.

computational costs for training and fine-tuning and requires the availability of a high-performance GPU server. SimBa 2023 achieves competitive scores, and its offline computations are relatively cheap, consisting only of computing embeddings using pre-trained language models and other textual features for the verified claims.

In order to answer RQ5, we now investigate the question whether the two best performing approaches, supervised RIET Lab and unsupervised SimBa 2023, are suitable for online fact-checking applications where response

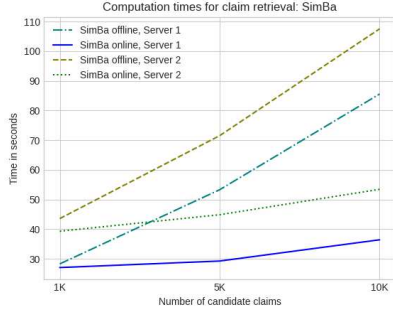
Level	System	Runtime	Constant costs
$O(1)$	ES Baseline	$C1 * f(1)$	$C1 = \text{ES look-up}$
$O(k)$	ClaimLinker	$C2 * f(1) + C3 * f(k)$ with $k = \max(30, D)$ and $C2 < C3$	$C2 = \text{ES look-up}$, $C3 = \text{pre-processing and textual similarity features}$
	TIET	$C4 * f(1) + C5 * f(k)$ with $k = \max(10000, D)$ and $C4 < C5$	$C4 = \text{ES look-up}$, $C5 = \text{embedding-based similarities}$
$O(D)^*$	RIET Lab	$C6 * f(D) + C7 * f(k)$ with $k = \max(25, D)$ and $C6 < C7$	$C6 = \text{inference using sentence transformer}$, $C7 = \text{GPTNeo-based reranking}$
	SimBa 2023	$C8 * f(D) + C9 * f(k)$ with $k = \max(50, D)$ and $C8 < C9$	$C8 = \text{embedding-based similarities}$, $C9 = 3 \text{ embedding-based similarities plus lexical text overlap}$
$O(D)$	UofSheffield	$C10 * f(D)$	$C10 = \text{combination of BM25 and TF-IDF scores, pre-processing, cosine similarities}$
	Check_square	$C11 * f(D)$	$C11 = \text{embedding-based similarities}$
	NLP&IR@UNED	$C12 * f(D)$	$C12 = \text{embedding-based similarities, pre-processing, textual similarity features, inference using neural network}$

Table 23: Runtime complexities. Complexity levels: $O(1)$ - Constant runtime, $O(k)$ - Linear runtime w.r.t. k (constant iff $k < D$), $O(D)^*$ - Linear runtime with reduced C = reduced slope, $O(D)$ - Linear runtime.

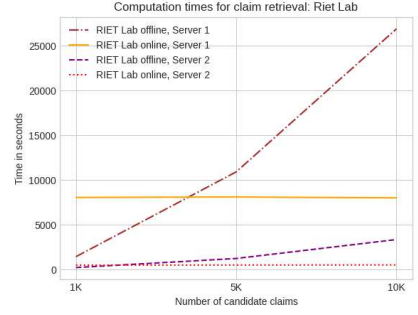
times matter. We experimentally measure their offline computation and online retrieval times.

Figure 8 shows the time required to link the set of 200 claims. Table 24 details the average time required to retrieve verified claims for one input claim, depending on dataset size, illustrating both total runtimes and scalability.

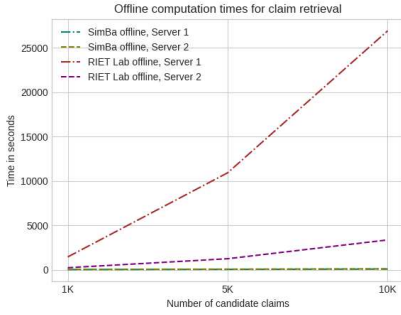
The offline computations for *RIET Lab* consist of training of the transformer-based candidate selection blocking step, and fine-tuning the GPT-Neo 1.3 Billion parameter model as reranker. For *SimBa 2023*, offline computations refer to the generation of three different embeddings plus tokenization for all verified claims. As shown in Figure 8a and Table 24, the transformer-based blocking step in *RIET Lab* is more efficient than *SimBa 2023*'s pair-wise comparisons regarding complexity, but *RIET Lab*'s constant costs are more than 200 times higher on Server1 and ~ 10 times higher on Server2 for a database size of 10k. The differences regarding the runtimes for Server1 vs. Server2 are due to the fact that *RIET Lab* benefits greatly from



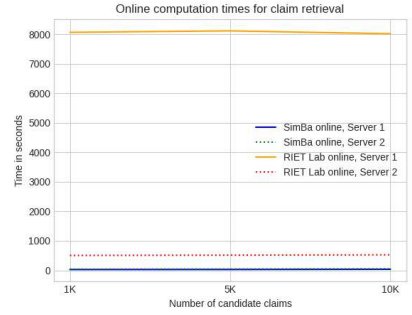
(a) Average retrieval runtime (in seconds) per input claim for different dataset sizes: SimBa 2023



(b) Average retrieval runtime (in seconds) per input claim for different dataset sizes: RIET Lab



(c) Average offline retrieval runtime (in seconds) per input claim for different dataset sizes



(d) Average online retrieval runtime (in seconds) per input claim for different dataset sizes

Fig. 8: Offline and online runtimes to retrieve targets for 200 input queries for *SimBa 2023* and *RIET Lab*.

Method	Hardware	Time (1K claims)	Time (5K claims)	Time (10K claims)
RIET Lab (offline)	Server 1	7.243	54.594	134.444
RIET Lab (online)	Server 1	38.659	39.651	41.270
SimBa 2023 (offline)	Server 1	0.141	0.266	0.428
SimBa 2023 (online)	Server 1	0.135	0.146	0.182
RIET Lab (offline)	Server 2	1.174	6.238	16.814
RIET Lab (online)	Server 2	2.533	2.579	2.631
SimBa 2023 (offline)	Server 2	0.218	0.358	0.538
SimBa 2023 (online)	Server 2	0.197	0.224	0.267
SimBa 2023 (offline)	Laptop	1.131	4.845	9.385
SimBa 2023 (online)	Laptop	0.324	0.335	0.380

Table 24: Average retrieval runtime per claim (in seconds) for different dataset (D) sizes.

GPUs, but the GPUs on Server1 do not meet the requirements regarding available memory and thus the computations were performed on CPUs only. *SimBa 2023*, on the other hand, is able to use all GPUs and CPUs on both servers for its computations. Offline computations (i.e. updates of the model) are very costly for *RIET Lab*, while for *SimBa 2023*, they can even be computed on a laptop. As Table 24 reveals, *SimBa 2023* is faster on a laptop than *RIET Lab* is on a high-performance GPU server. Online retrieval times for *SimBa 2023* are fast enough for online applications on all hardware setups. For *RIET Lab*, a server with a high-capacity GPU is required.

6.4 Discussion

To answer RQ1, our data analysis in Section 5 reveals that the benchmark datasets for the claim retrieval task are characterized by a high rate of token overlaps, which explains the relatively high performance of baseline methods relying on BM25 and other lexical similarity scores, as well as the dominance of embedding-based methods (cf. Section 6). As the benchmarks are built using real data, they contain noise, such as very long and very short claims, partly due to extraction errors. As this reveals vulnerabilities of methods regarding overfitting and allows to better measure their robustness, we argue that this actually makes the data more useful to assess the suitability of methods in real-world systems, where they will be suspected to this kind of noise as well. Data from different discourse types, tweets, and political debates, are similar regarding characteristics such as ratio of token overlaps. The increased task difficulty for the debates data can to a large part be attributed to the lacking context information in its input queries, a challenge that would typically not arise for our use-case, a real-world claim retrieval system with queries entered by users. Therefore, these different datasets are not very well suited to investigate the general generalizability of models across different data distributions. The tweets datasets are limited in the sense that there is always one matching verified claim for each input query, while in real systems drawing on multiple fact-checking portals, we can expect anything from zero to multiple potential matching verified claims, since different fact-checking portals may check the same check-worthy claims. The debates data loosens these restrictions, allowing multiple matching targets. However, this data contains input queries that are part of check-worthy claims, but do not represent check-worthy claims themselves and are not the type of input that is expected in claim retrieval applications. Therefore, training and testing models on this data may not give insights into their ability to detect the essential, check-worthy part of verified claims, but instead limits the task to detecting overlapping parts in input queries and verified claims. With this, the data does not allow to distinguish methods that may differ regarding this aspect. However, as users are not expected to enter any non-check-worthy claims, this may not impede the performance of models trained on this data. Finally, whether the high ratio of overlaps and the small impact of using the more abstractive title information

can be attributed to data selection biases or is inherent to the claim retrieval task, remains unclear.

Investigating RQ2, we find that the highest performance is achieved by supervised methods on all datasets but *2020-tweets* (cf. Tables 19 and 20). The best-performing approach overall is transformer- and GPTNeo-based *RIET Lab*, which achieves up to 0.956 MAP@5 on *2022-tweets*, followed by the unsupervised *SimBa 2023*, which achieves the top score on *2020-tweets* with 0.962. While supervised/fine-tuned systems also generally dominated the leaderboards of the CLEF CheckThat! Lab challenges, their performance turned out to fluctuate considerably between and within different datasets and is hard to predict, e.g. with *RIET Lab* performing ~ 0.1 points worse on *2020-tweets* than on the other tweet datasets, with individual runs ranging from 0.801 to 0.957 in MAP@5 and *UofSheffield*'s performance diverging by ~ 0.5 points between their *2020-tweets* and *2021-tweets* models (cf. Tables 19), despite the similarity of the datasets (cf. Section 4). As our cross-evaluation results in Tables 21 and 22 show, this difference is caused by what the model learned rather than by differences in the test sets. This and our analyses showing that noisy data may greatly bias the systems' performance (cf. Tables 12 and 13) highlight the importance of the availability of large amounts of training data to avoid overfitting, especially in the presence of noisy data. Acquiring such data, however, is costly, and so are training and fine-tuning using large language models. Unsupervised methods reach a similar or even better performance, with *SimBa 2023* achieving the top rank on the *2020-tweets* data with a MAP@5 score of 0.964 and most unsupervised methods outperforming most supervised ones, except *RIET Lab*, on the debates datasets (cf. Tables 17 and 18). *SimBa 2023* is the best unsupervised method, ranking second behind *RIET Lab* on all datasets except *2020-tweets*. On this dataset, it achieves the top score (cf. Tables 14, 15, 16, 17, 18).

Regarding RQ3, our results show that supervised methods for this task generalize well across the used datasets of different discourse types, i.e. models learned on tweets data work well for retrieving claims from debates and vice versa. However, as the models are prone to overfitting, their performance depends heavily on the training data and fluctuates strongly within the same discourse type (cf. Tables 21 and Tables 22): E.g. *NLP&IR@UNED* and *UofSheffield* learned their most useful model on *2020-tweets* which then yields the best results on all datasets, including those featuring claims from political debates. In general, training models on the *2020-tweets* dataset yields the best-performing models for most systems. This is the dataset where token overlaps are the most indicative to distinguish matching and non-matching pairs, the variance in input query length is highest in the training set and lowest in the test set. As the datasets are relatively small, however, supervised methods may overfit their models: e.g. *RIET Lab*'s worst and best model trained on *2020-tweets* yields a MAP@5 score of 0.801 vs. 0.957, respectively. The debates data features fewer positive training examples than the tweets data, shorter inputs (i.e. less information), and the imbalance of negative vs. positive examples is

pronounced. Training on this data yields the lowest performance. Finally, the cross-evaluation confirms that not only the training data for the debates data is more difficult, but also the test sets are harder than those of the tweets datasets: models trained on the tweets data that perform well across datasets still yield lower scores on the debates test datasets. These have a higher number of non-matching verified claims to choose from, multiple correct matches, the input queries contain less words and characters than the tweet input queries and are less self-contained (cf. Section 5).

Section 6.3, Figure 8 and Table 24 answer RQ4: methods relying on an index structure naturally exhibit constant low retrieval time. The use of blocking techniques can render approaches using costly computations competitive in regards to efficiency without yielding bad performance (cf. Table 19).

Efficiency can be greatly improved by performing costly computations for all claims in the dataset offline, storing them for comparisons with new input claims, and computing the costly features only for the latter. In combination with blocking, this can cap and reduce the retrieval time of approaches making heavy use of costly features. We show that even though the best approach *RIET Lab*'s online retrieval has higher efficiency than the second best approach *SimBa 2023*'s in terms of run time complexity, the absolute costs of the former are too high to allow its application for online claim retrieval. Also, its offline computational costs are high and it requires the availability of a high-end server both for training and inference. Thus, to answer RQ5, our analysis suggests that the unsupervised *SimBa 2023* is well-suited for online fact-checking applications, with very low requirements regarding hardware as it can even be run on a laptop, while its performance is close to *RIET Lab*'s (+0.108, -0.011, -0.008, -0.039, -0.019 on *2020-tweets*, *2021-tweets*, *2022-tweets*, *2021-debates* and *2022-debates*, respectively), and, as outlined above, generally more stable. From this, we conclude that costly fine-tuning and supervision of neural models is not necessary for the task of retrieving fact-checked claims and that an unsupervised method such as *SimBa 2023* that exploits pre-trained language models such as Sentence-T5 and MPNet is overall the best choice for (online) claim retrieval systems.

7 Conclusions and Future Work

We presented an analysis of benchmark datasets for the claim retrieval task and an evaluation of the performance of different supervised and unsupervised approaches on different data to compare their robustness and scalability and extract insights on which methods work best and why.

We showed that an unsupervised system that exploits the complementarity of different embeddings achieves state-of-the-art performance on par with the best supervised method and outperforms all others while being highly efficient, scalable and robust.

Our study has several limitations: we focused on the retrieval performance for the English language. In future work, it would be interesting to compare the performance when applied to different languages and in multilingual settings.

Moreover, the usefulness of different language models was shown to vary considerably. It remains unclear which properties exactly render a model useful for this task and how biases introduced by language models (cf. [15]) affect claim retrieval, e.g. yielding higher performance for certain topics or claims expressing stances uttered with preference by advocates of particular political leanings.

Thus, as part of our future work, we plan to investigate in more detail the different semantics that are encoded in different language models and their informative value and biases for this task.

Despite the fact that the datasets belonging to the same discourse type, tweets vs. political debates, are continuous extensions of their previous versions, we found relatively large fluctuations in systems' performance scores when using either of them. At the same time, the debate data differs from the tweets data in more aspects than the genre and features input queries that do not match the task definition entirely. Regarding characteristics such as the importance of word overlaps, the tweets and debates datasets are similar. Thus, our comparative analysis provides insights on the performance of the different methods on different datasets and their robustness, but cannot provide insights on the generalizability of methods regarding application on different discourse types or out-of-distribution data more generally. Relating to this, our data analysis investigated several characteristics that have been recognized as important factors in previous works and in a top-down fashion drawing on domain insights. Yet, the data may contain additional biases introduced by the methodology of dataset creation, which may not be captured by our analysis [8] and which may artificially reduce task difficulty [53]. Finally, the data diverges from data that can be expected in real-world claim retrieval systems in the sense that it does not feature ranked lists with multiple similar claims verifying an input query, as can be expected when drawing on data from multiple fact-checking portals. Thus, our study lays the groundwork to understand the state-of-the-art in claim retrieval and put existing methods into perspective, but further research is needed to fully understand task performance in real-world systems.

To tackle these issues, we aim to build on this study and collect real user-generated queries and relevance assessments by integrating different claim retrieval methods into the *ClaimLinker* application to link real input queries to claims from multiple fact-checking portals, enhance the existing benchmark data and more directly evaluate the usefulness of generated predictions.

Declarations

Ethical approval

Not applicable

Competing interests

The authors have no competing interests to declare that are relevant to the content of this article.

Authors' contributions

K.B., S.D., P.F., and K.T. conceived the idea of the paper. K.B. performed the data analysis steps, prepared the figures, modified the code of the compared systems to enable the replication studies, performed their evaluation and time measurements and wrote the main manuscript text. A.H. developed and extended the SimBa approach, generated replication data for SimBa, the ES Baseline and assisted in cross-validating some of the findings. U.M. implemented the extension of the ClaimLinker tool, guided mainly by P.F., assisted by K.B., and generated data for ClaimLinker and TIET. S.D., P.F., A.H. and K.T. contributed to the experimental design and the manuscript text. All authors reviewed the manuscript.

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Availability of data and materials

All data including the source code to replicate this study are available on GitLab³⁸

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